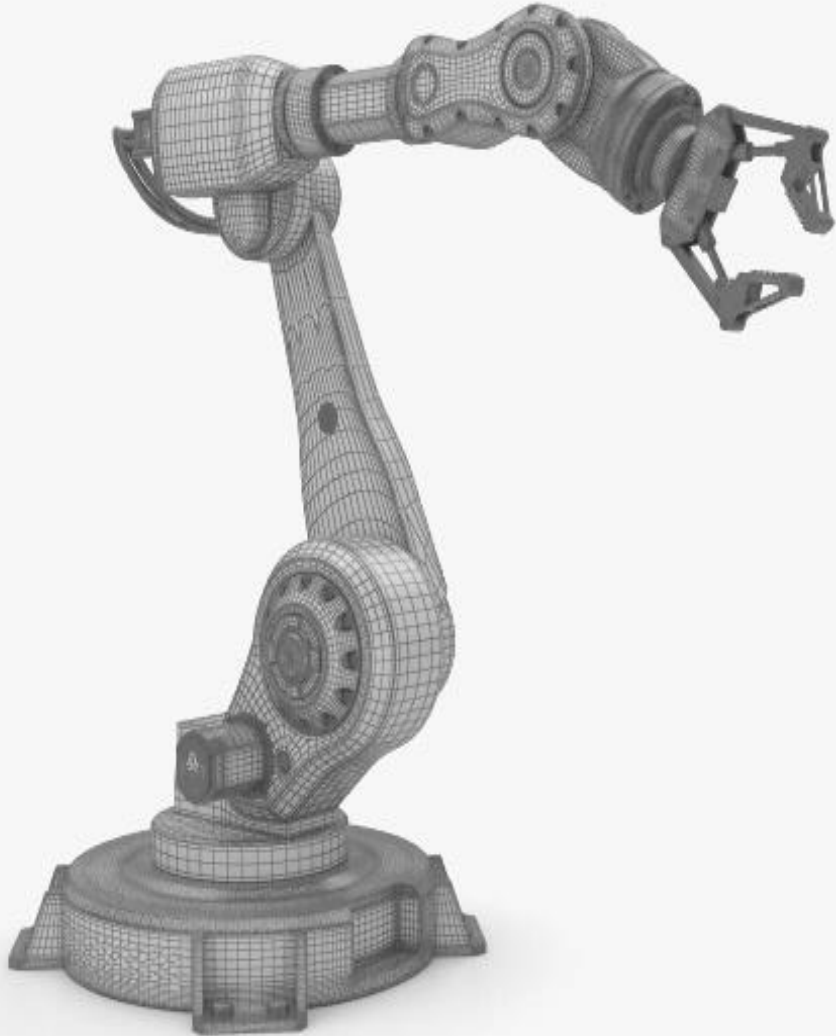


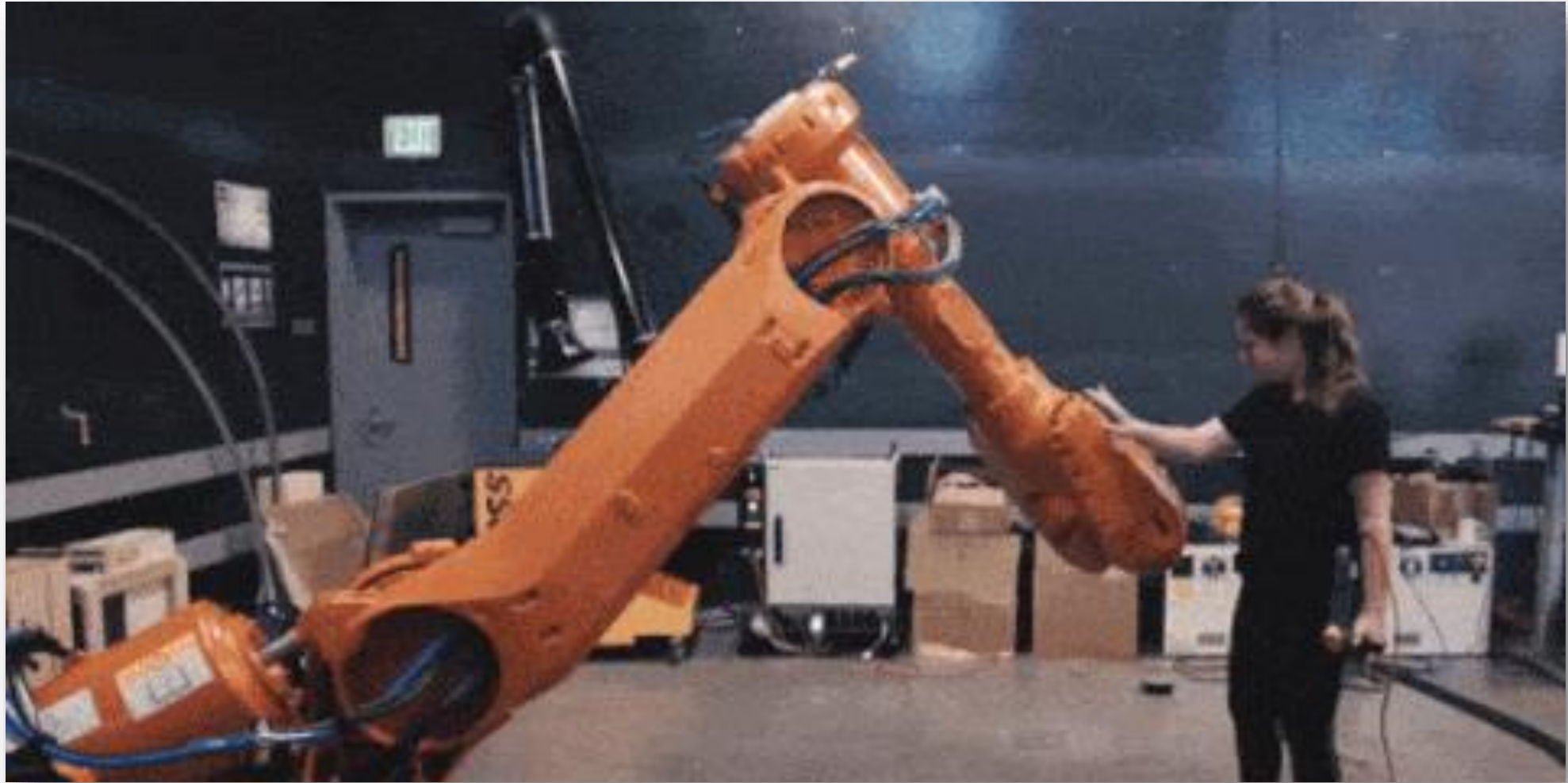


Robotic Systems

Chapter III Rigid Body motion

Prof. Dr. Ibrahim Hamarash
University of Kurdistan-Hewlêr

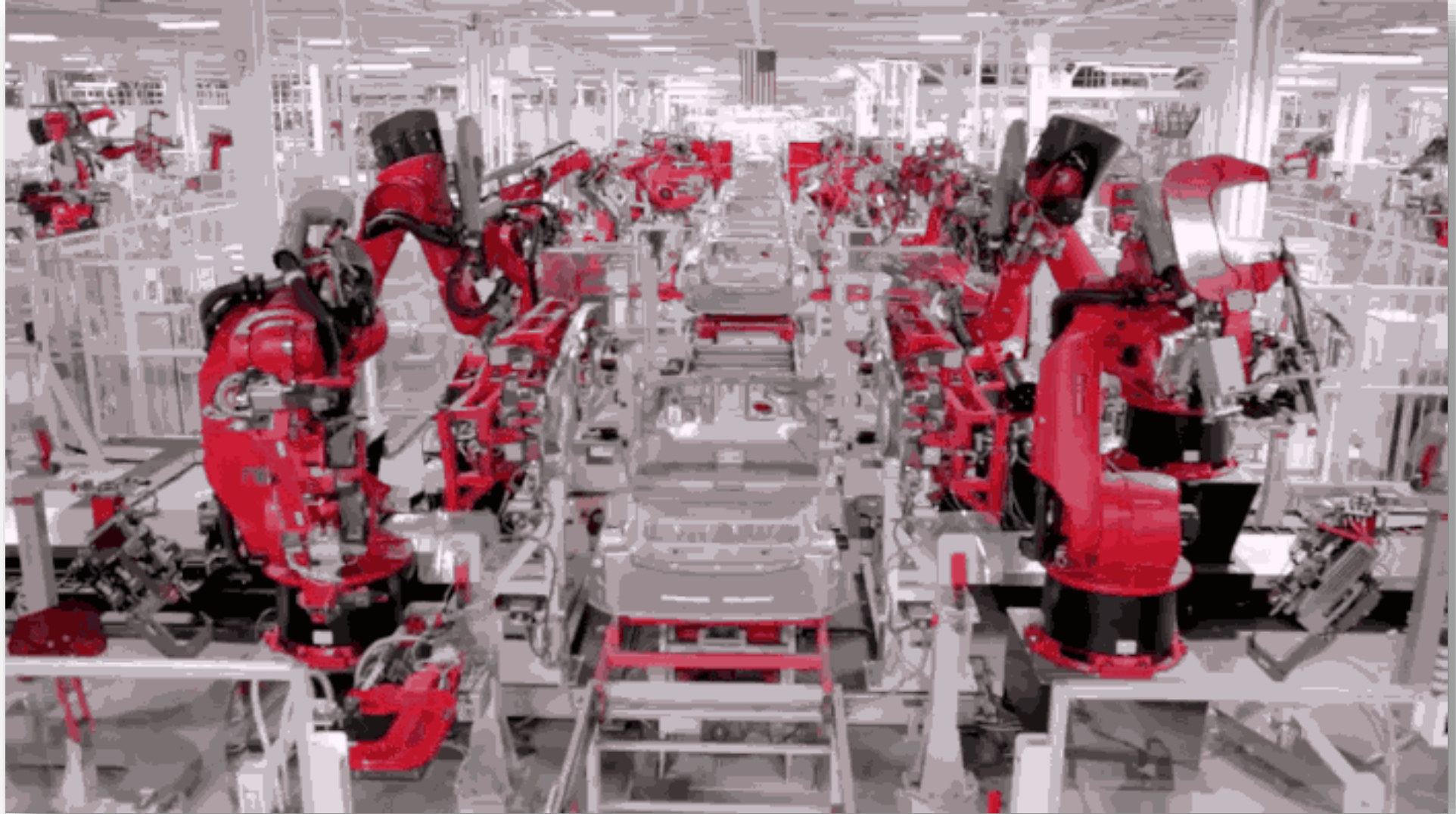
Rotation and Displacement



Rotation and Displacement



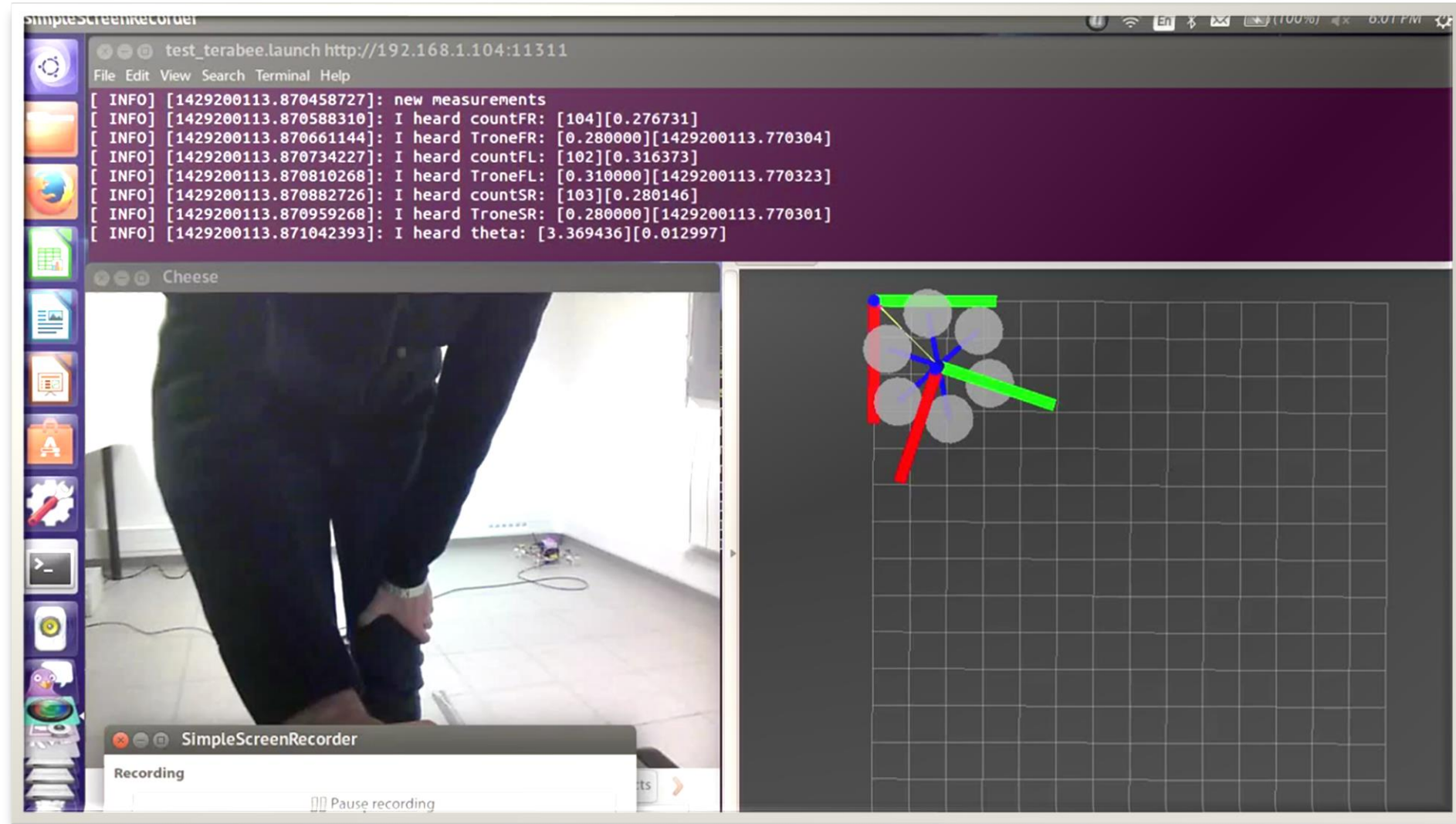
Rotation and Displacement



Rotation and Displacement

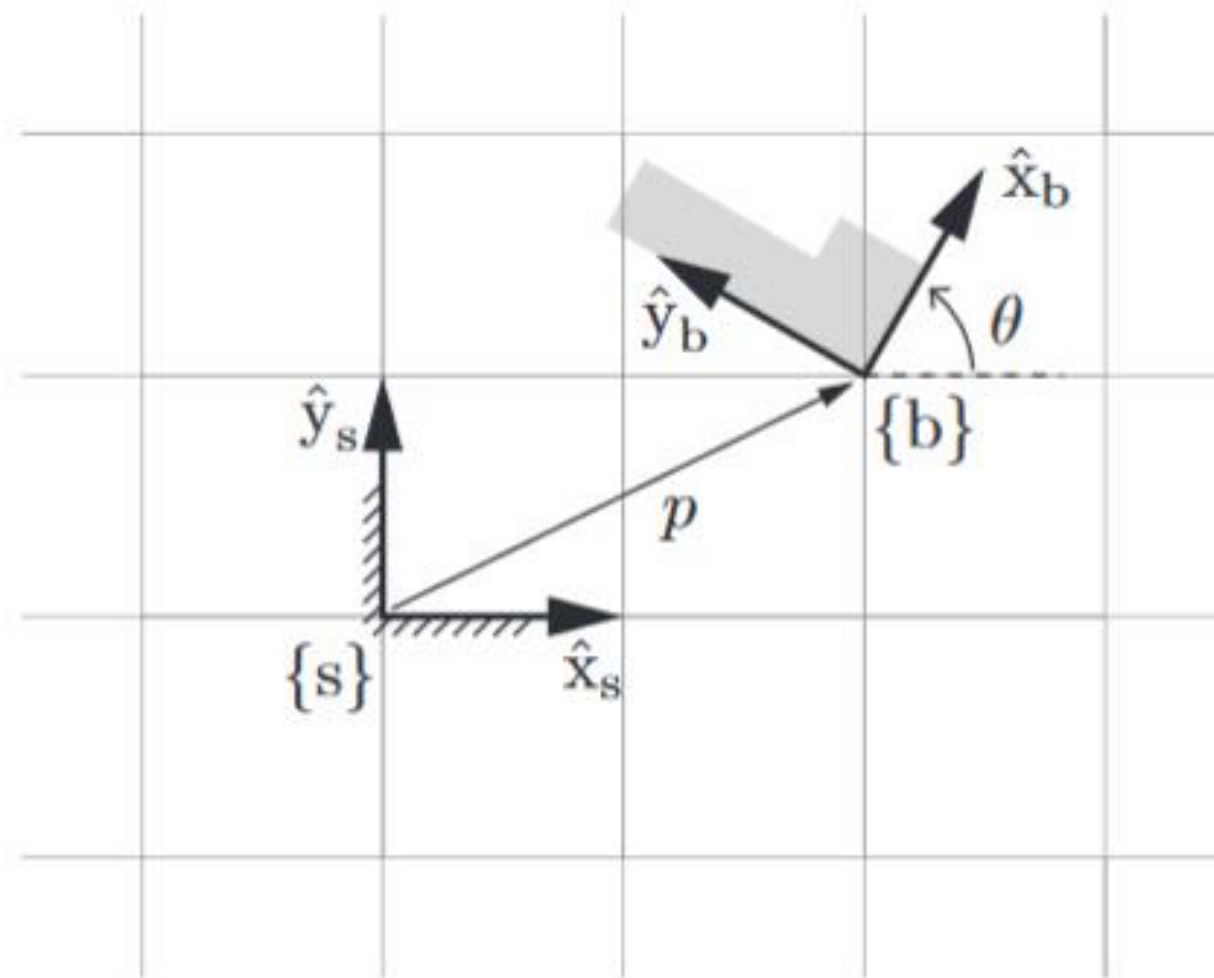


Rotation and Displacement



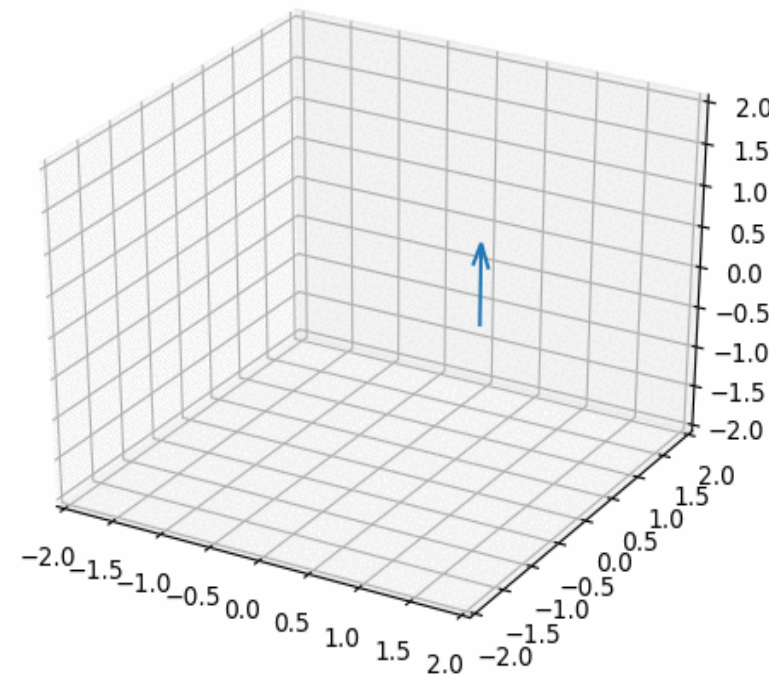
Position and Orientation

A fundamental requirement in robotics is to represent the position and orientation of objects in an environment. Such objects include robots, cameras, workpieces, obstacles and paths.



Rotation and Displacement in 3D

A minimum of six numbers is needed to specify the position and orientation of a rigid body in three-dimensional physical space.

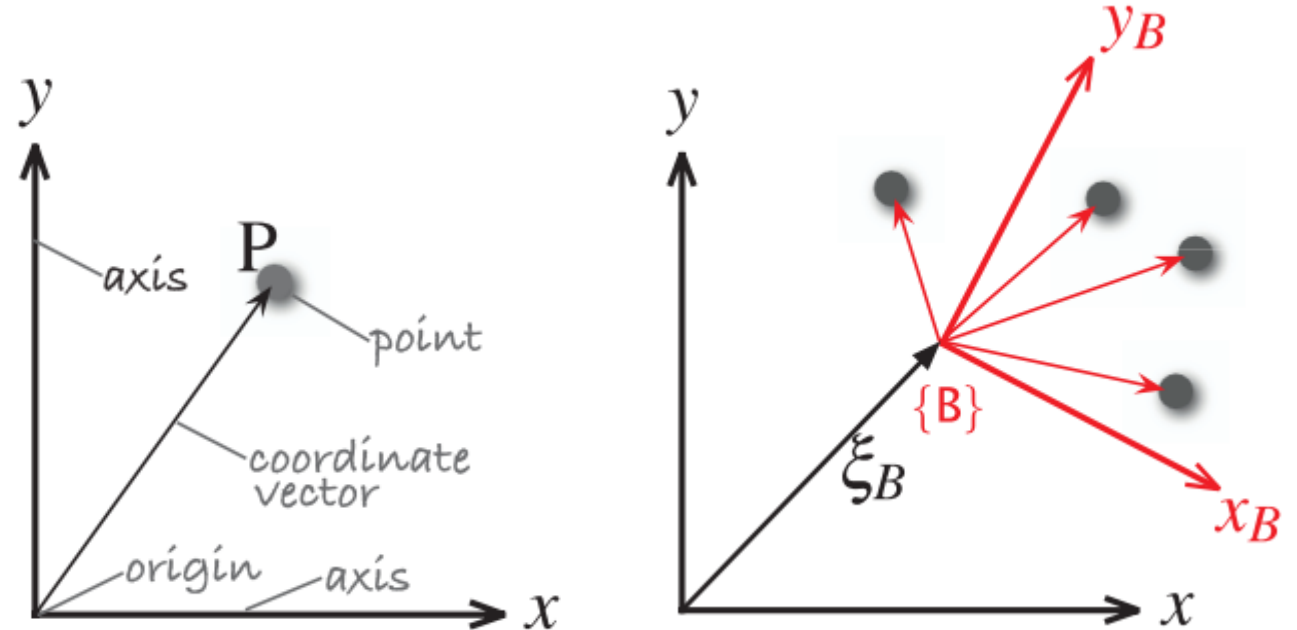


Reference frame and Body frame

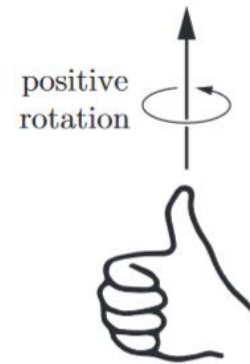
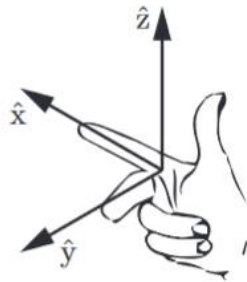
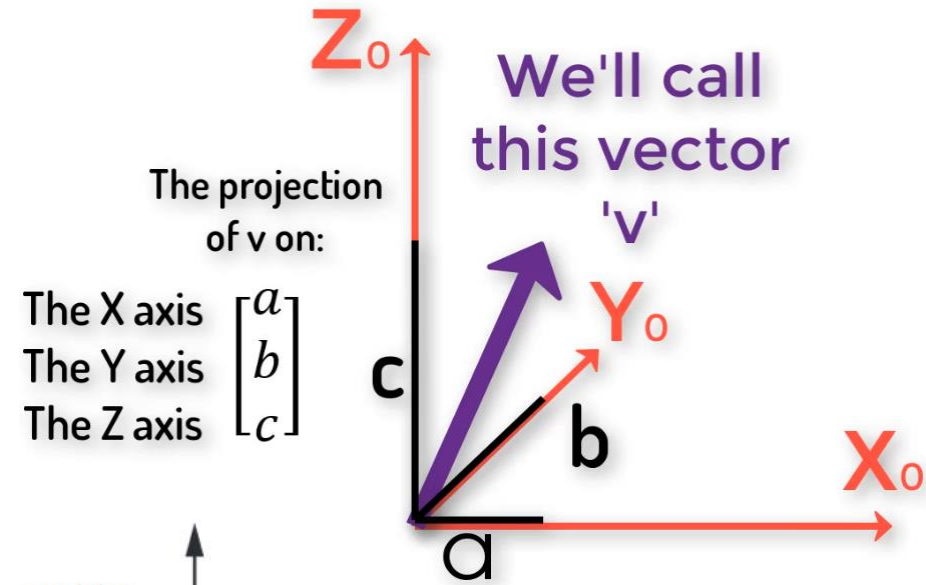
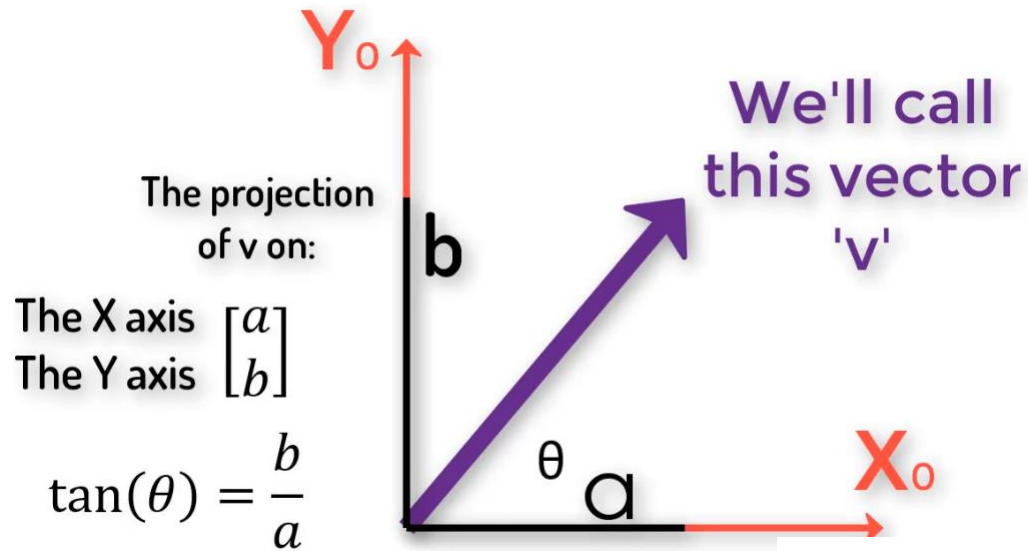
We need to consider a set of points that comprise some object.

We assume that the object is *rigid* and that its constituent points maintain a constant relative position with respect to the object's coordinate frame.

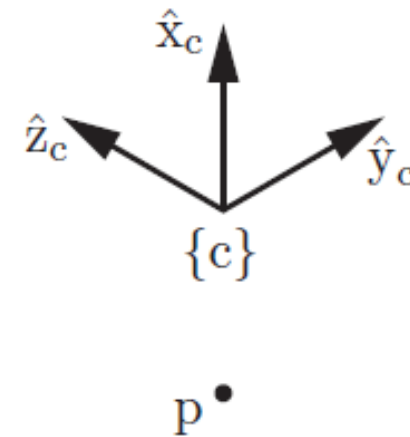
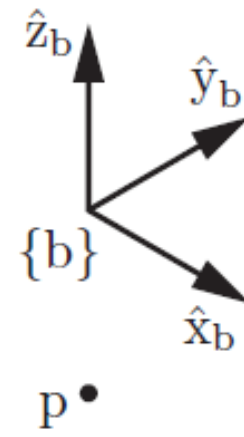
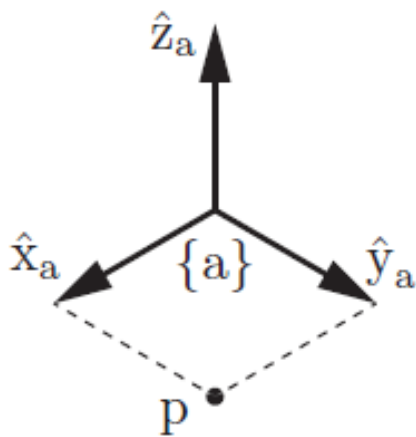
Instead of describing the individual points we describe the position and orientation of the object by the position and orientation of its coordinate frame. A coordinate frame is labelled, $\{B\}$ in this case, and its axis labels X_B and Y_B adopt the frame's label as their subscript.



Vector geometry



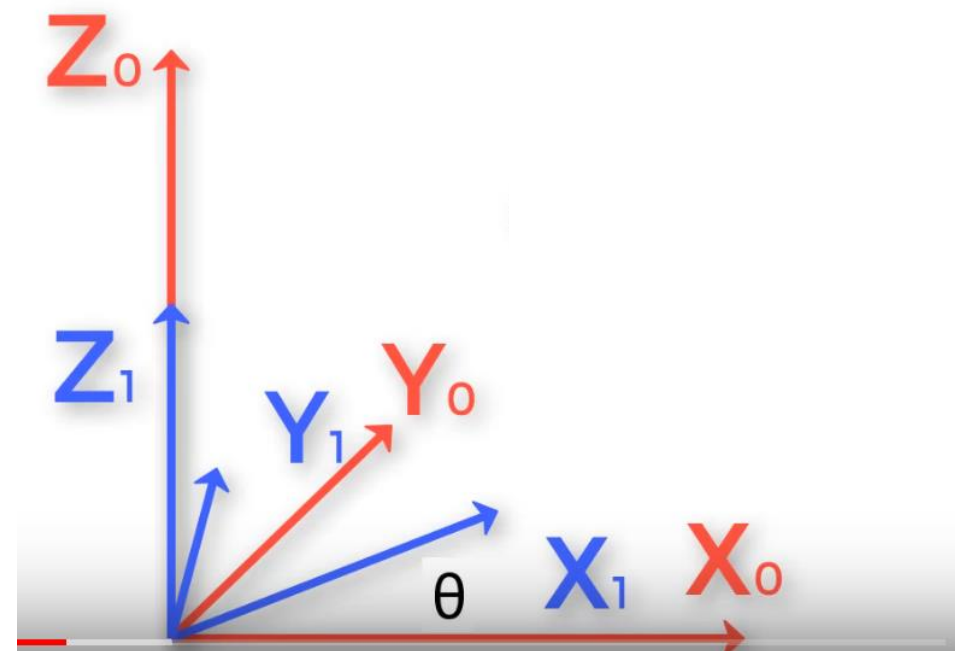
Ex. The same space and the same point p represented in three different frames with different orientation. Fixed frame s is aligned with the frame a .



$$R_a = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_b = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_c = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

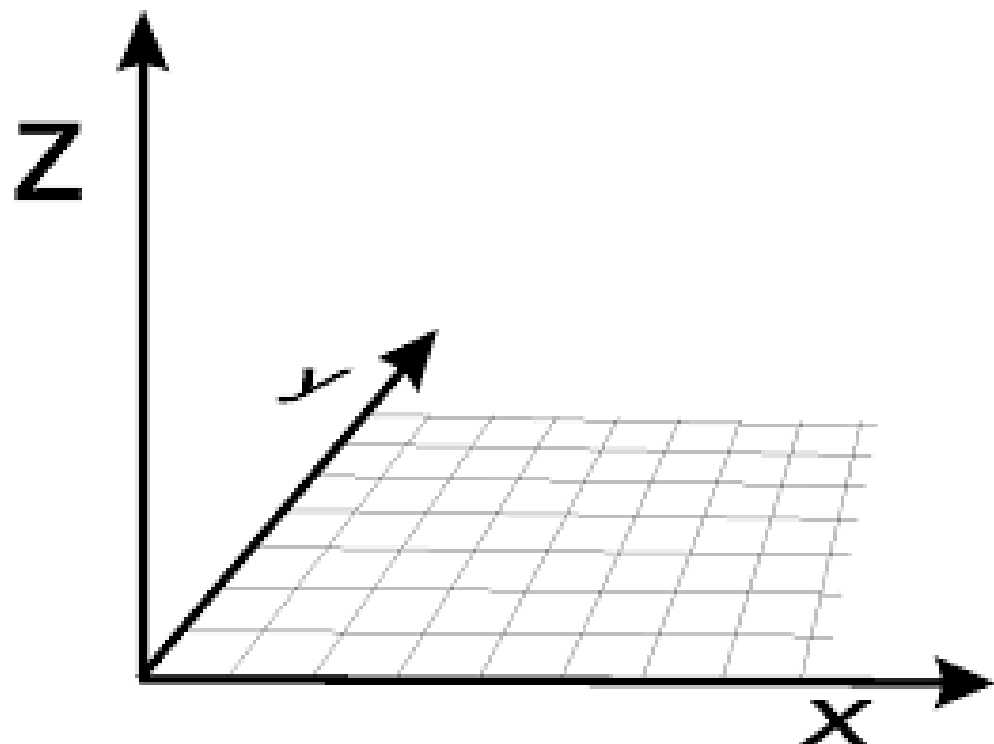
Rotation Matrix

$$R_{01} = \begin{matrix} & \text{on:} & X_1 & Y_1 & Z_1 \\ X_0 & & \square & \square & \square \\ Y_0 & & \square & \square & \square \\ Z_0 & & \square & \square & \square \end{matrix}$$



$$R_Z = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The three rotation matrices



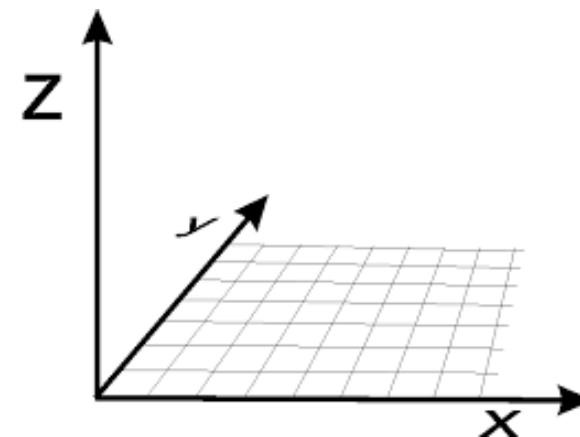
$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The three rotation matrices

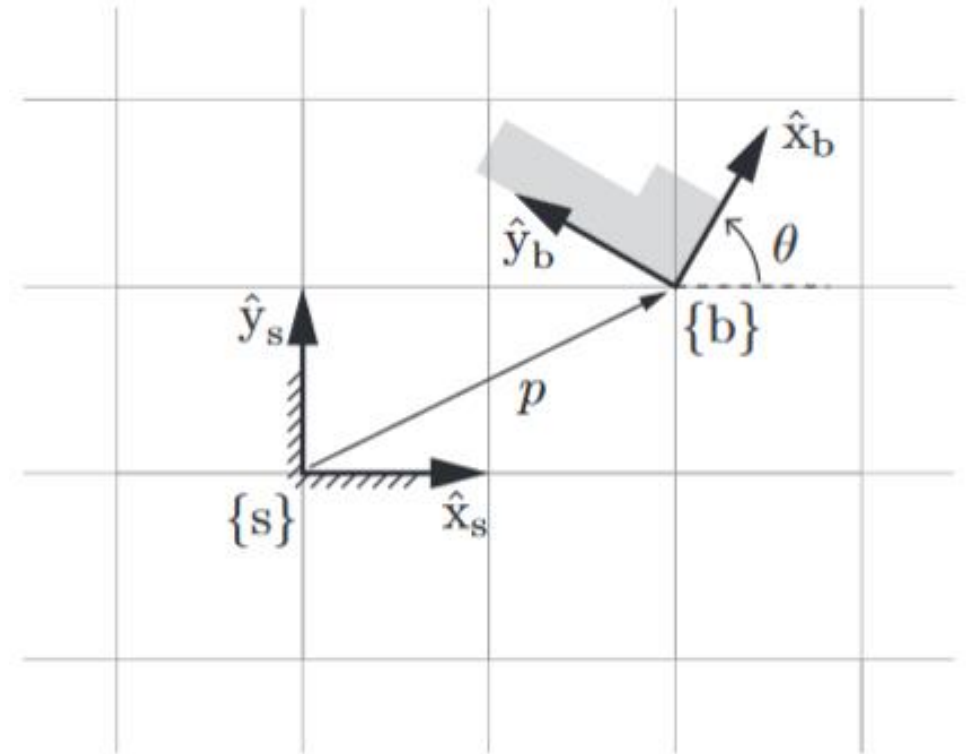
Rot(x,θ) representation:



$$\text{Rot}(\hat{x}, \theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}, \quad \text{Rot}(\hat{y}, \theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$\text{Rot}(\hat{z}, \theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Example: Consider the planar body (the gray shape) shown in the figure; its motion is confined to the plane. Suppose that a length scale and a fixed reference frame $\{s\}$ have been chosen as shown, with unit axes $\hat{x}_s$ and $\hat{y}_s$. Similarly, we attach a reference frame with unit axes $\hat{x}_b$ and $\hat{y}_b$ to the planar body. Because this frame moves with the body, it is called the body frame and is denoted $\{b\}$.



Properties of the rotation matrix

Consider a rotational matrix R

$$R = [\hat{x}_b \quad \hat{y}_b \quad \hat{z}_b] = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

1. The unit norm condition, $\hat{x}_b, \hat{y}_b, \hat{z}_b$ are all unit vectors

$$r_{11}^2 + r_{21}^2 + r_{31}^2 = 1,$$

$$r_{12}^2 + r_{22}^2 + r_{32}^2 = 1,$$

$$r_{13}^2 + r_{23}^2 + r_{33}^2 = 1.$$

Properties of the rotation matrix

2. The orthogonality condition $\hat{x}_b \cdot \hat{y}_b = \hat{x}_b \cdot \hat{z}_b = \hat{y}_b \cdot \hat{z}_b = 0$ (the $\cdot$ Denotes the inner product of the matrix)

$$\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$r_{11}r_{12} + r_{21}r_{22} + r_{31}r_{32} = 0,$$

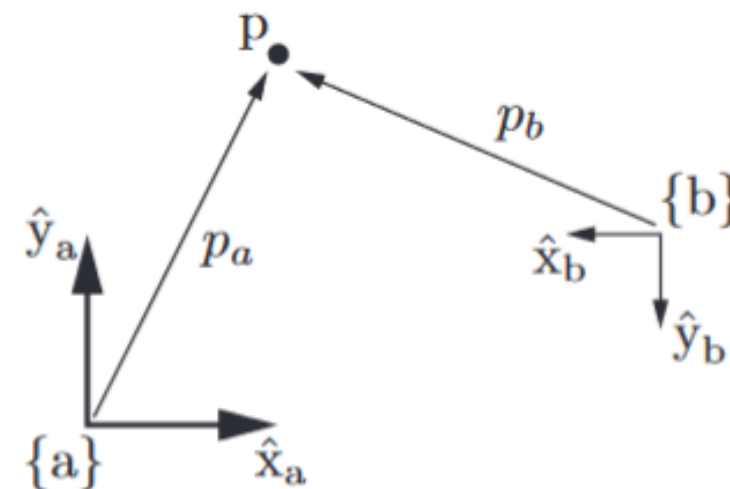
$$r_{12}r_{13} + r_{22}r_{23} + r_{32}r_{33} = 0,$$

$$r_{11}r_{13} + r_{21}r_{23} + r_{31}r_{33} = 0.$$

Or, $R^T R = I$

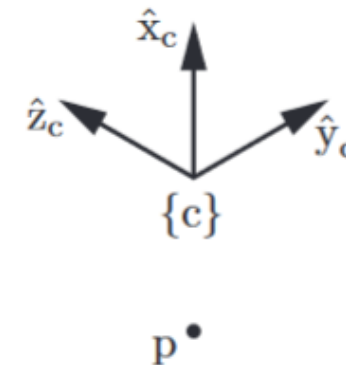
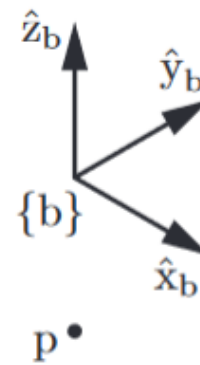
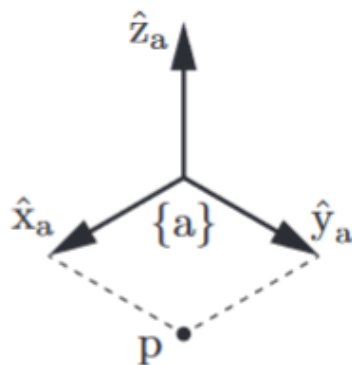
Uses of Reference Frames

1. Representing an Orientation
2. Changing the reference frame
3. Rotating a vector or a frame



$$p_a = (1, 2), p_b = (4, -2)$$

Example: Representing an Orientation



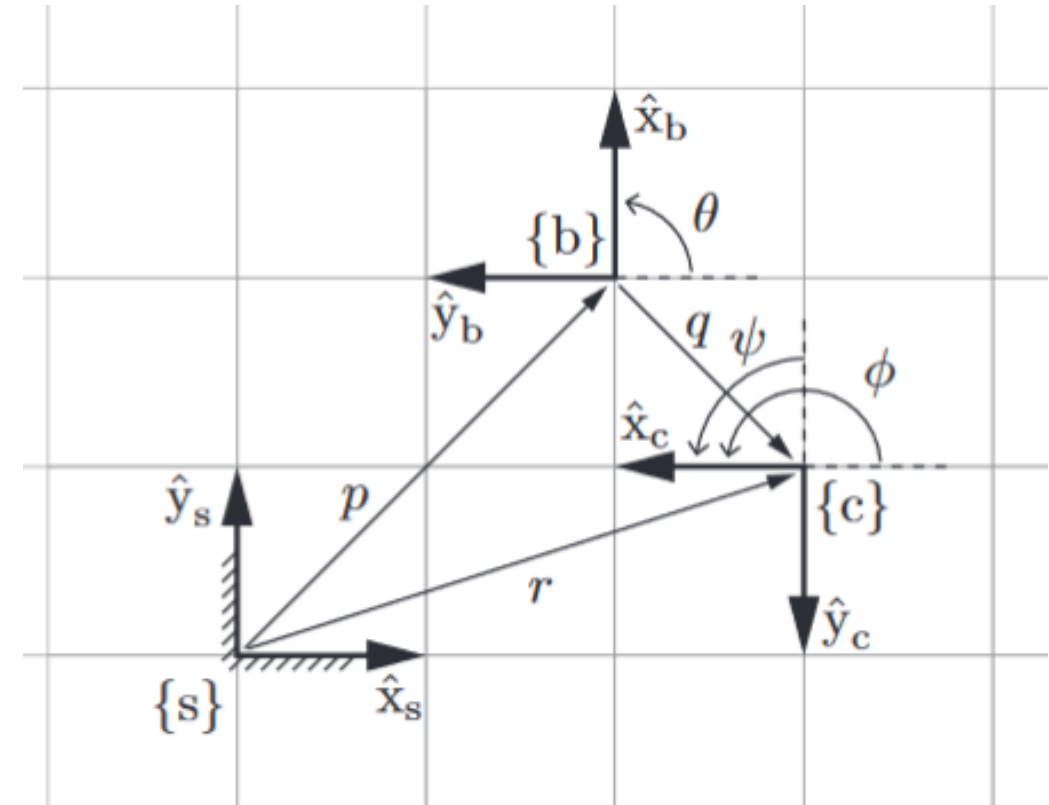
$$R_{ac} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}, \quad R_{ca} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}. \quad p_a = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad p_b = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad p_c = \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix}.$$

Note: If there is no possibility of confusion regarding the frames involved, we may simply write **R**.

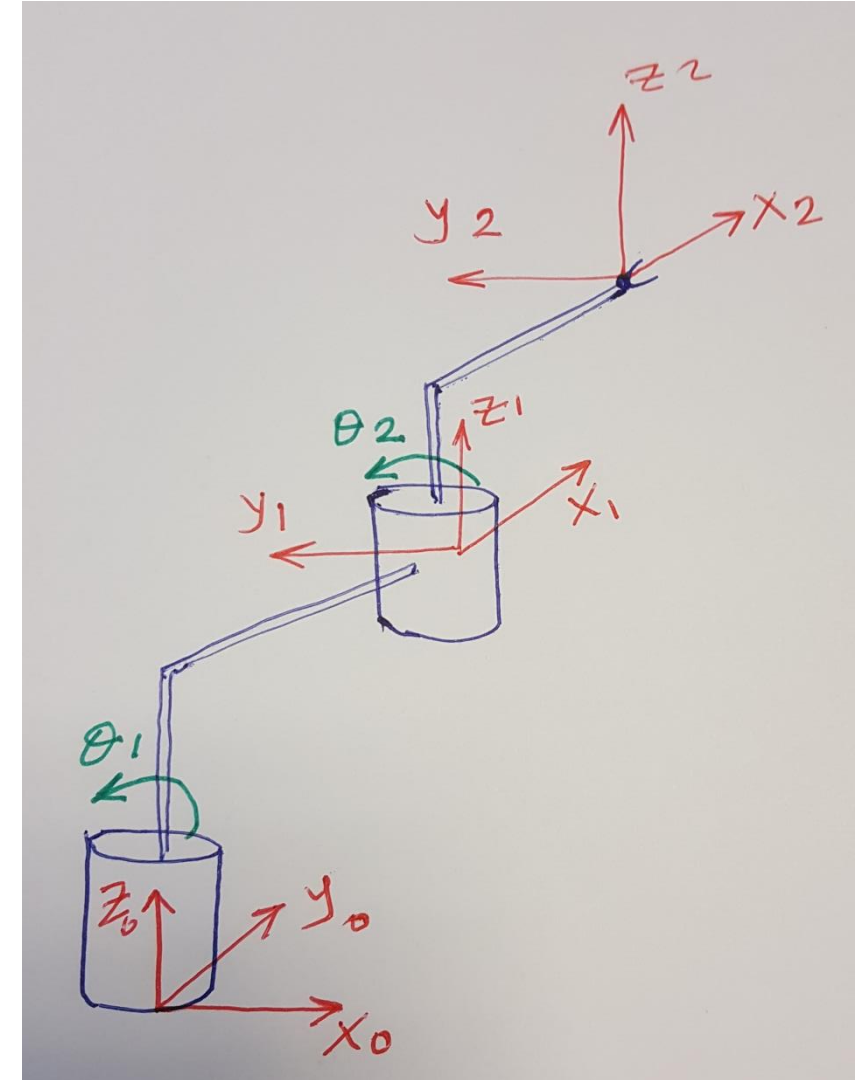
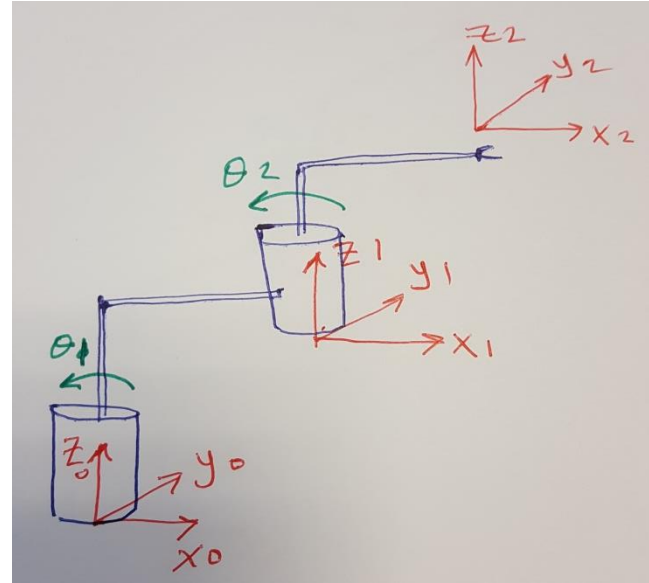
Example: Changing the reference frame

changing reference from $\{b\}$ to $\{s\}$

$$R_{sb}R_{bc}=R_{sc}$$



Example: Find the rotation matrix from the frame 2 to frame 0

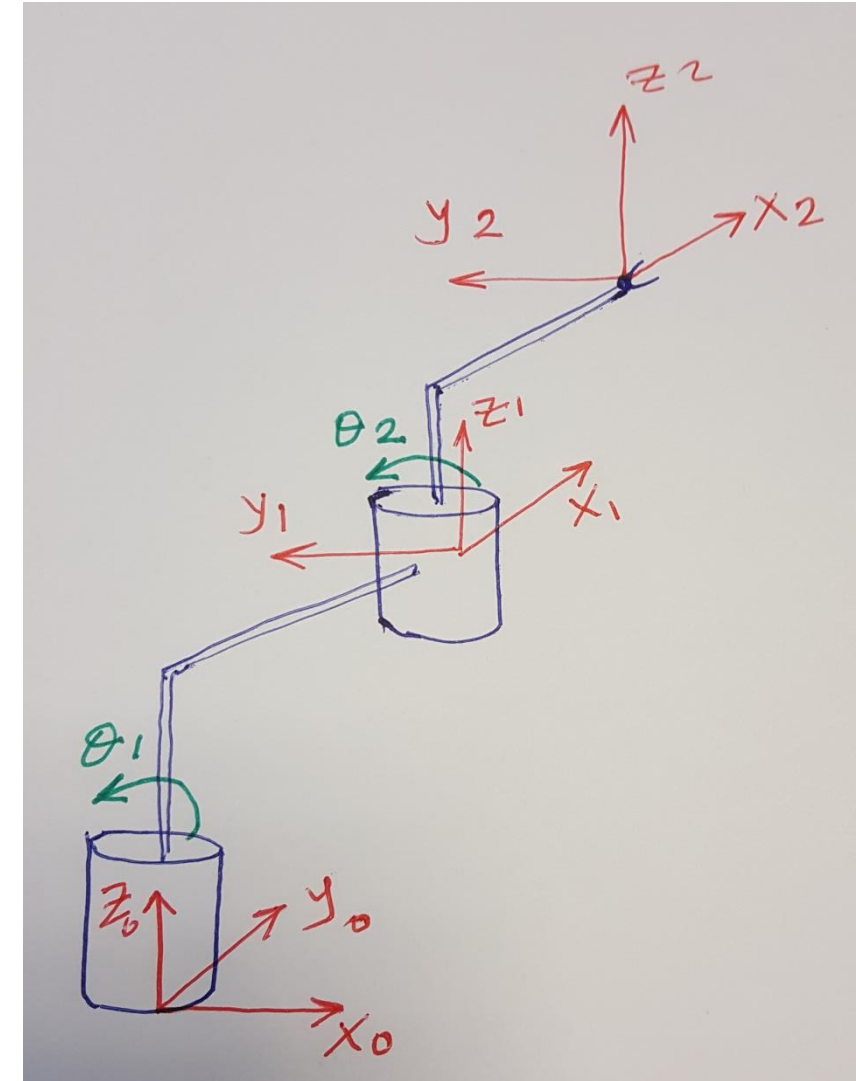


Rotation and Displacement

$$R_{01} = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_{01} = I \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

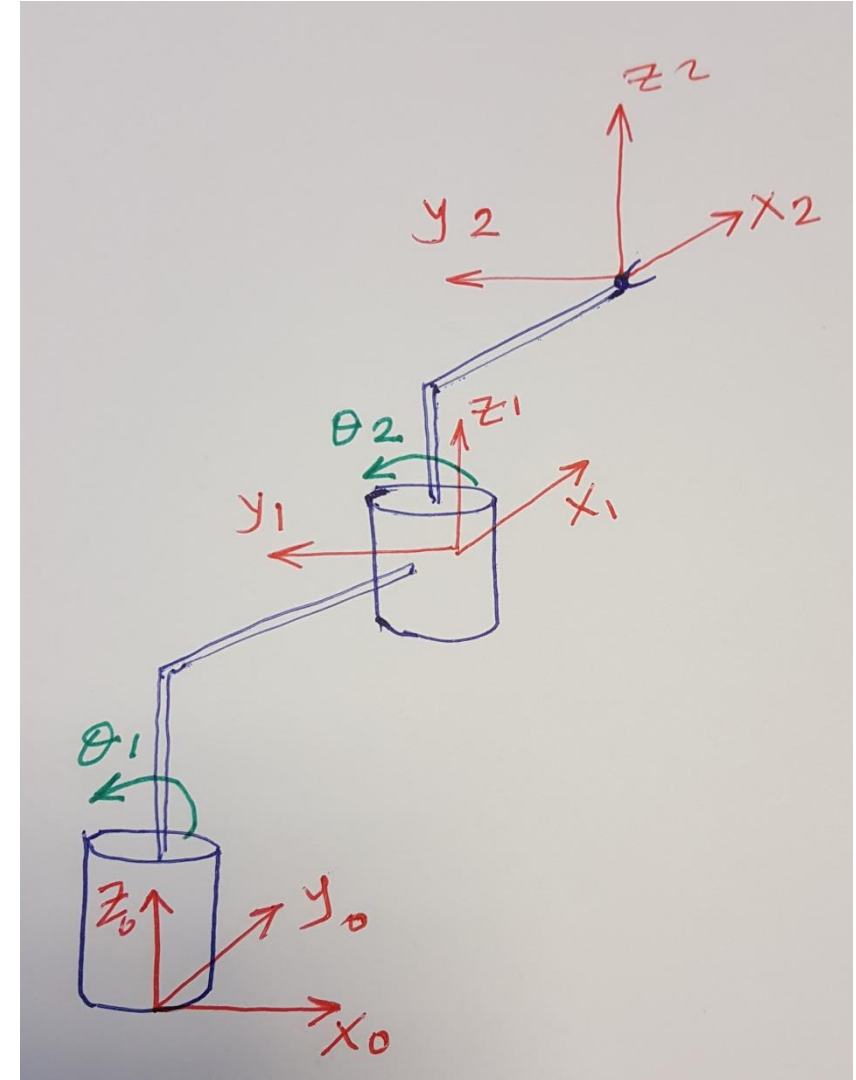


Rotation matrix

$$R_2^1 = I \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & 0 \\ \sin \theta_2 & \cos \theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

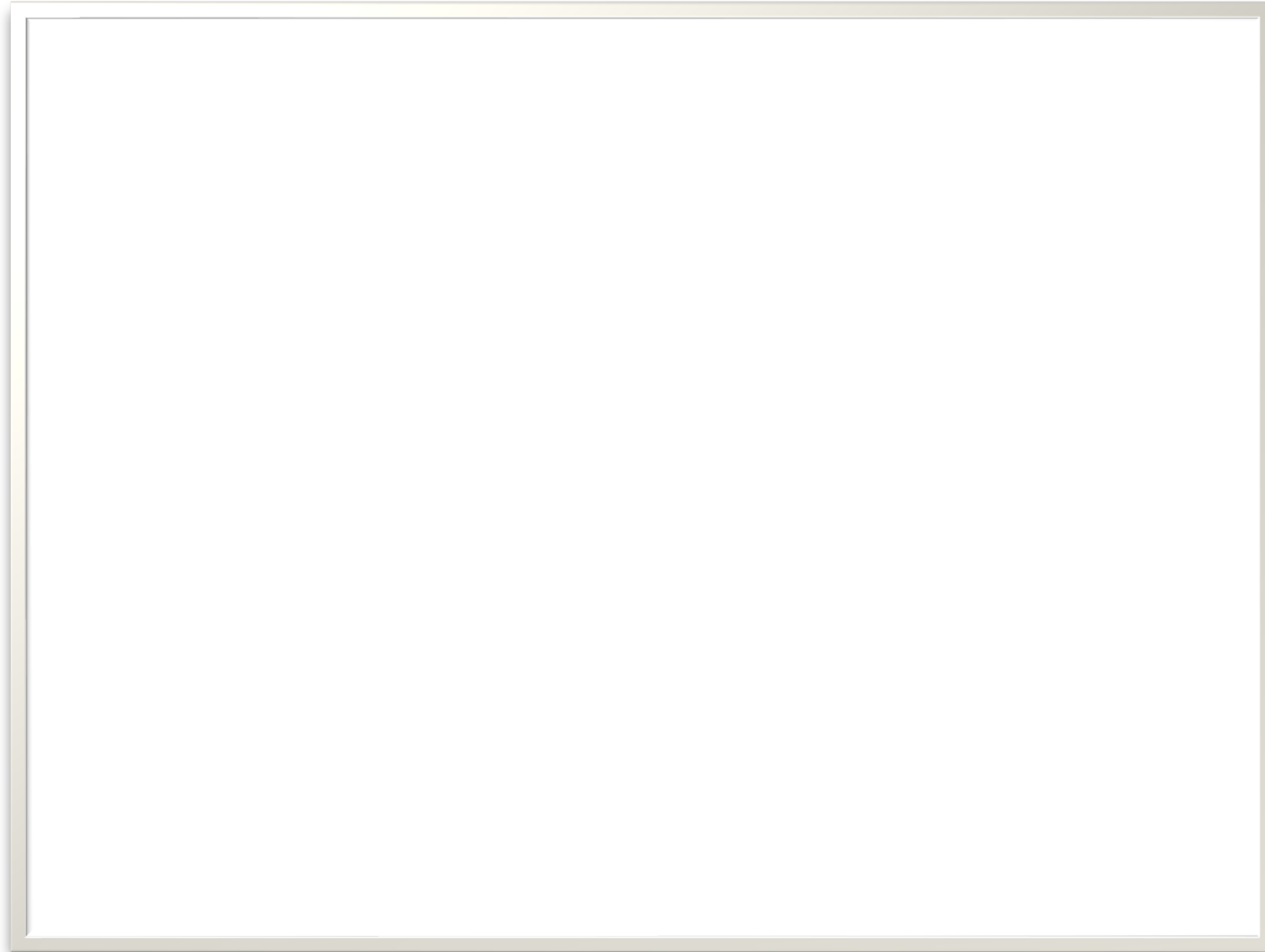
$$R_2^0 = R_1^0 R_2^1$$



Rotation matrix

$$R_{02} = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & 0 \\ \sin \theta_2 & \cos \theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Translation or Displacement



Translational Transformation

Let the translational displacement is given by the vector $\mathbf{d}$,

$$\mathbf{d} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k},$$

$\mathbf{d}$ can be described in a matrix form as,

$$\mathbf{V} = \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Using H for displacement an arbitrary vector q for a distance d is obtained by

$$\mathbf{q} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = [x \ y \ z \ 1]^T$$

$$\mathbf{v} = \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x+a \\ y+b \\ z+c \\ 1 \end{bmatrix}$$

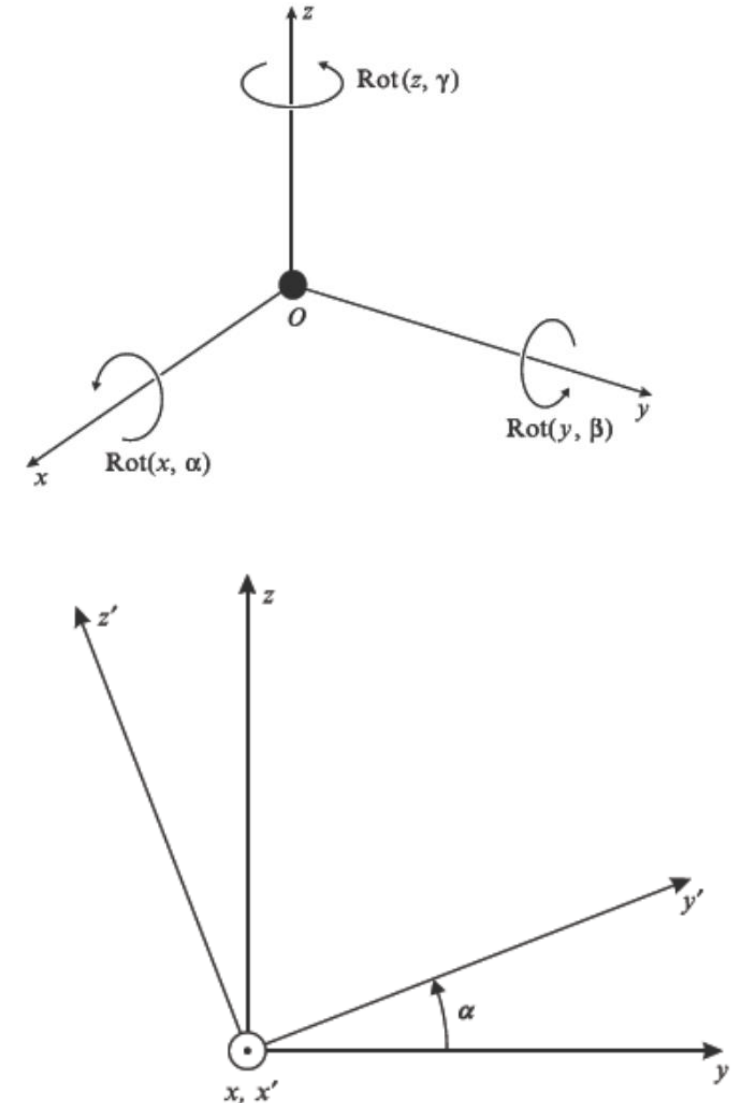
Example:

The vector $2\mathbf{i}+3\mathbf{j}+2\mathbf{k}$ is translationally displaced for distance $4\mathbf{i}-3\mathbf{j}+7\mathbf{k}$

$$\mathbf{v} = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 9 \\ 1 \end{bmatrix}$$

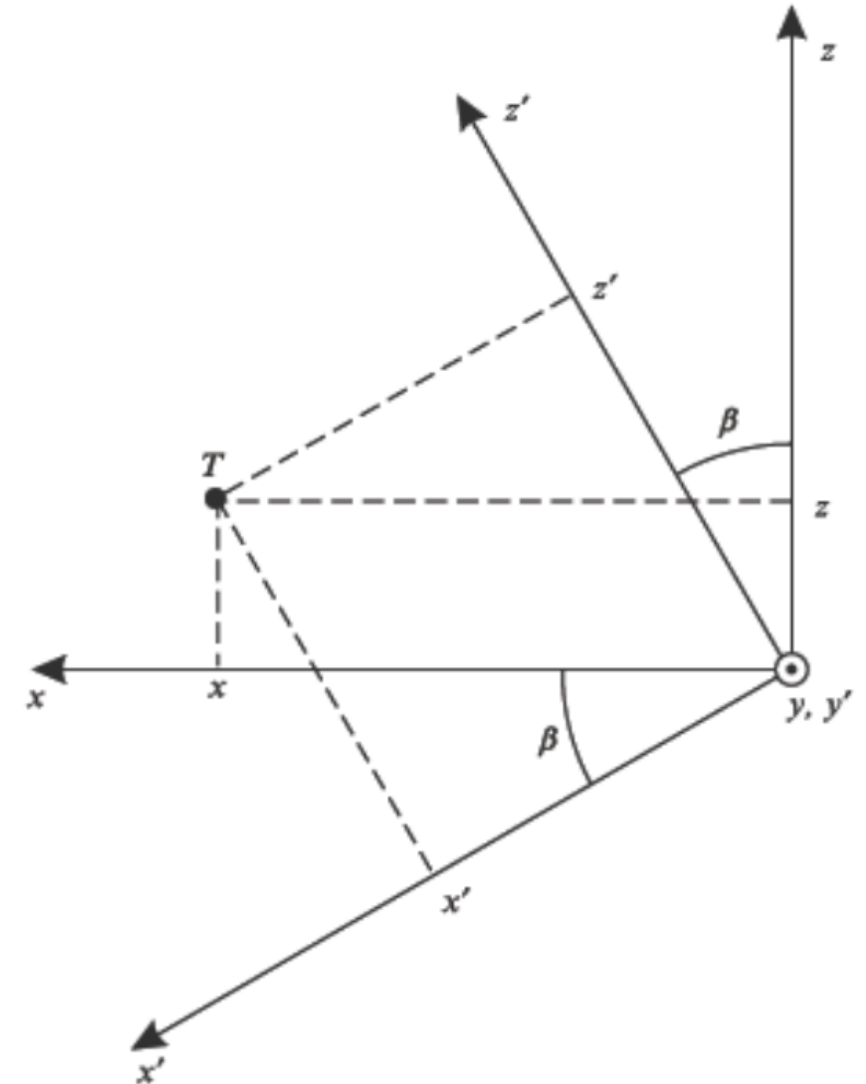
Example of Rotational Transformation

$$\text{Rot}(x, \alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha & 0 \\ 0 & \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Example: Rotation about y axis

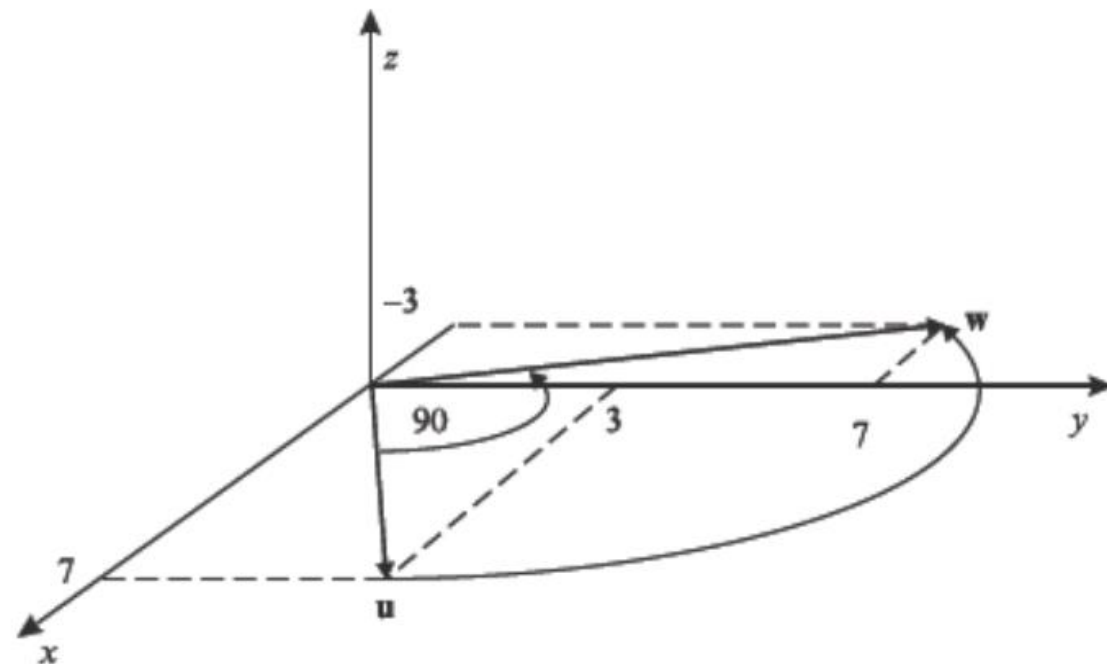
$$Rot(y, \beta) = \begin{bmatrix} x' & y' & z' \\ \cos \beta & 0 & \sin \beta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \beta & 0 & \cos \beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} x \\ y \\ z \end{matrix}$$



Example: Determine the vector w which is obtained by rotating the vector $u=7i+3j+0k$ for 90° in the counter clockwise around z -axis.

$$Rot(z, \gamma) = \begin{bmatrix} \cos \gamma & -\sin \gamma & 0 & 0 \\ \sin \gamma & \cos \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

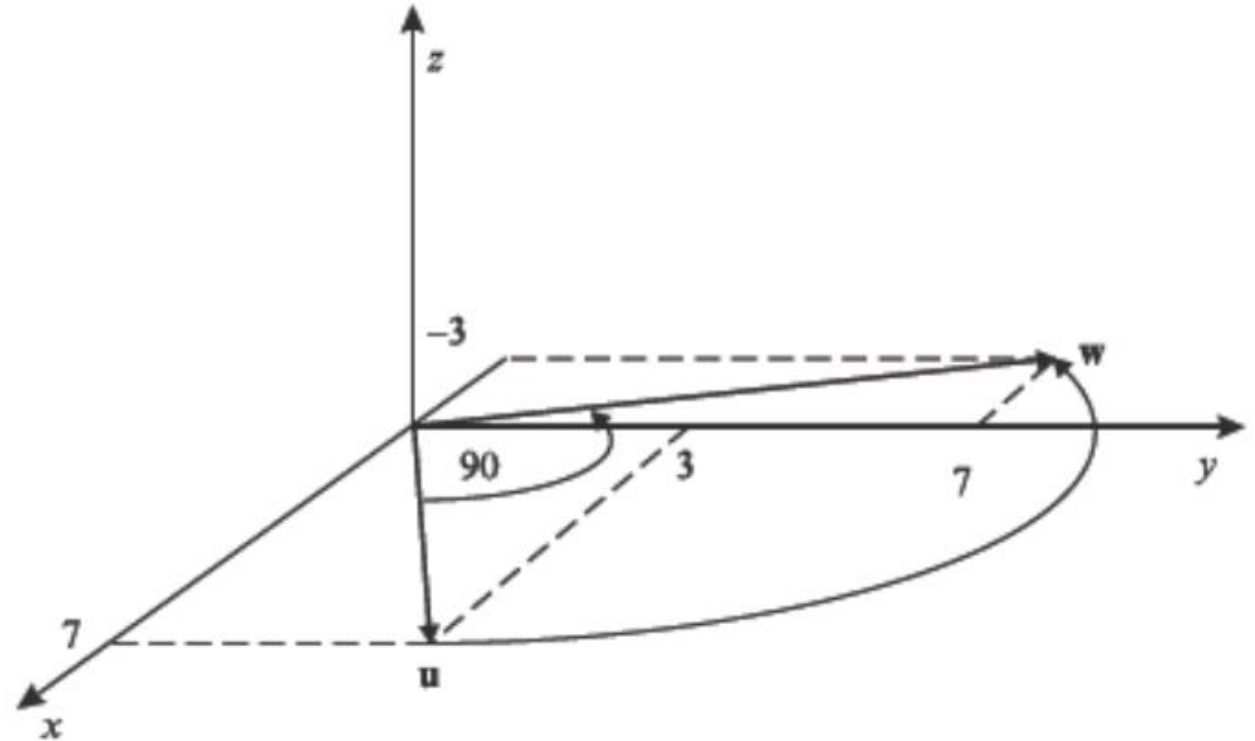
$$Rot(z, 90) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Example, cont.

$$W = \text{Rot}(z, 90) \times u$$

$$W = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 7 \\ 0 \\ 1 \end{bmatrix}.$$

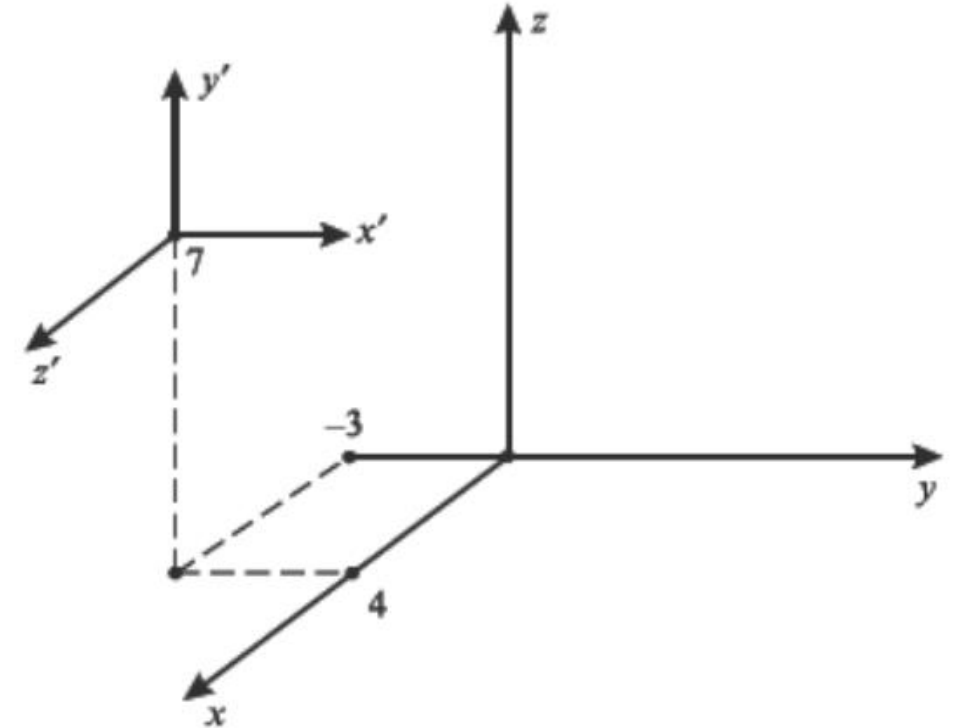


Transformation (Homogenous Rotation and translation) (Pose)

$$\mathbf{H} = \text{Trans}(4, -3, 7) \text{Rot}(y, 90^\circ) \text{Rot}(z, 90^\circ)$$

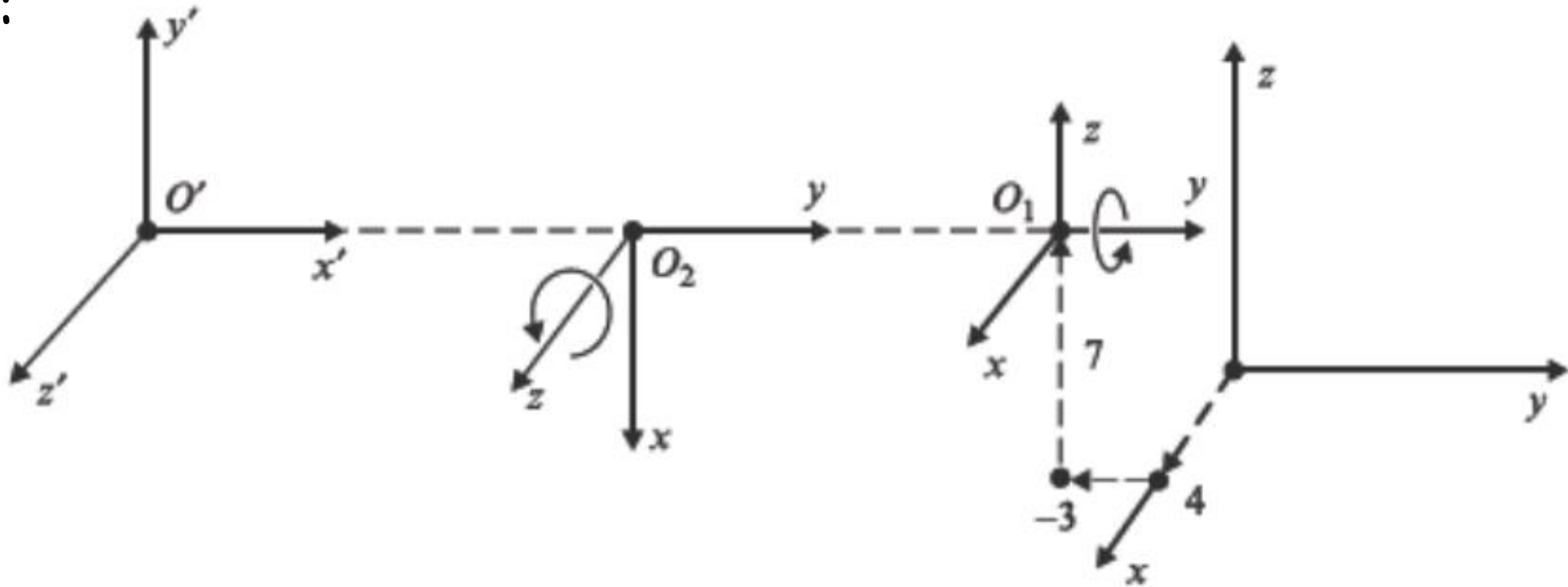
$$= \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 & 4 \\ 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$



$$\begin{bmatrix} [0] & [0] & [1] & 4 \\ 1 & 0 & 0 & -3 \\ [0] & [1] & [0] & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Example:



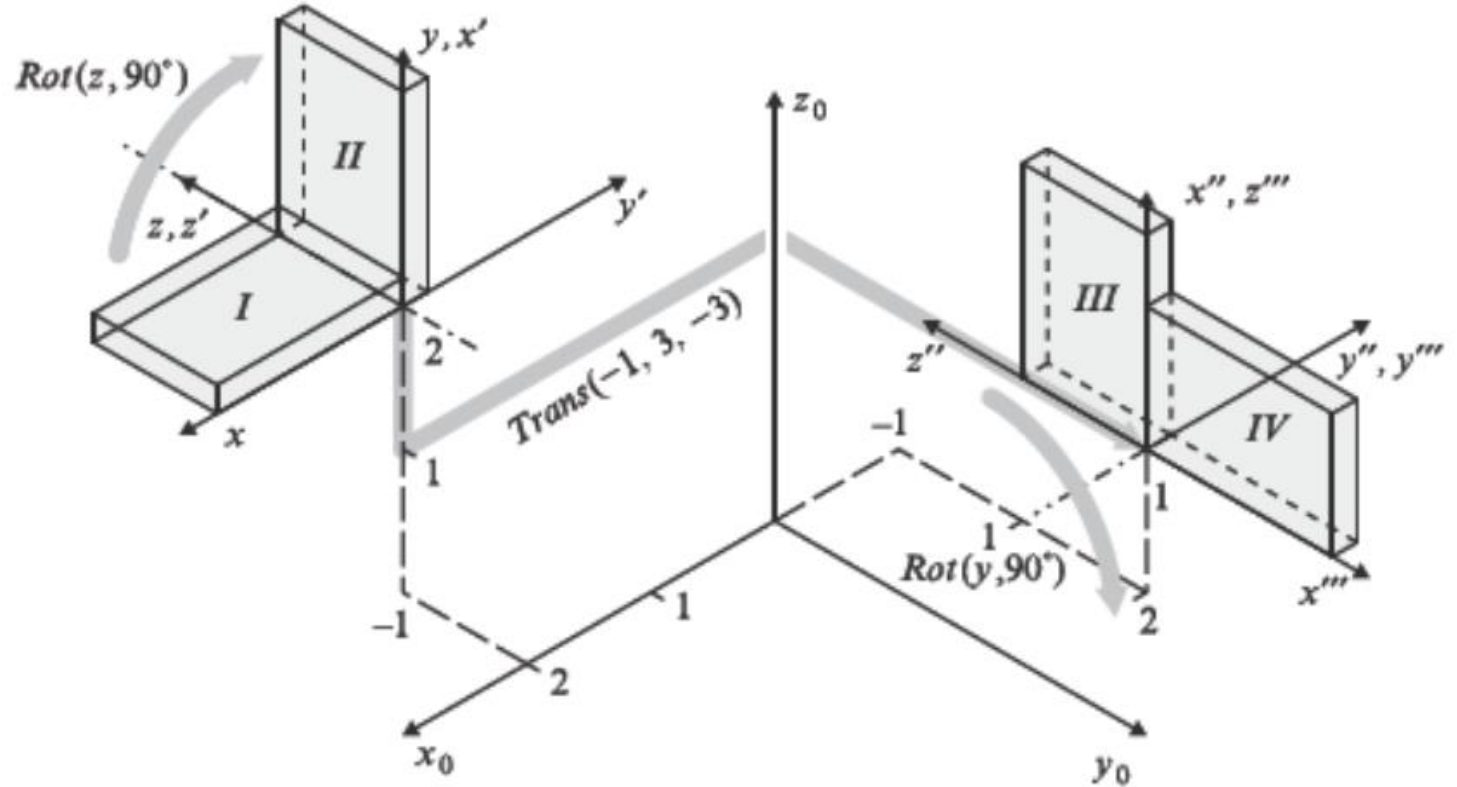
$$\mathbf{H} = \text{Trans}(4, -3, 7) \text{Rot}(y, 90^\circ) \text{Rot}(z, 90^\circ)$$

Example:

First, we wish to rotate the coordinate from x, y, z for 90° in the counterclockwise direction around z axis,

then displace for -1 along x' axis, 3 units along y' axis and -3 along z' axis.

Finally rotate for 90° around y'' axis in the positive direction.

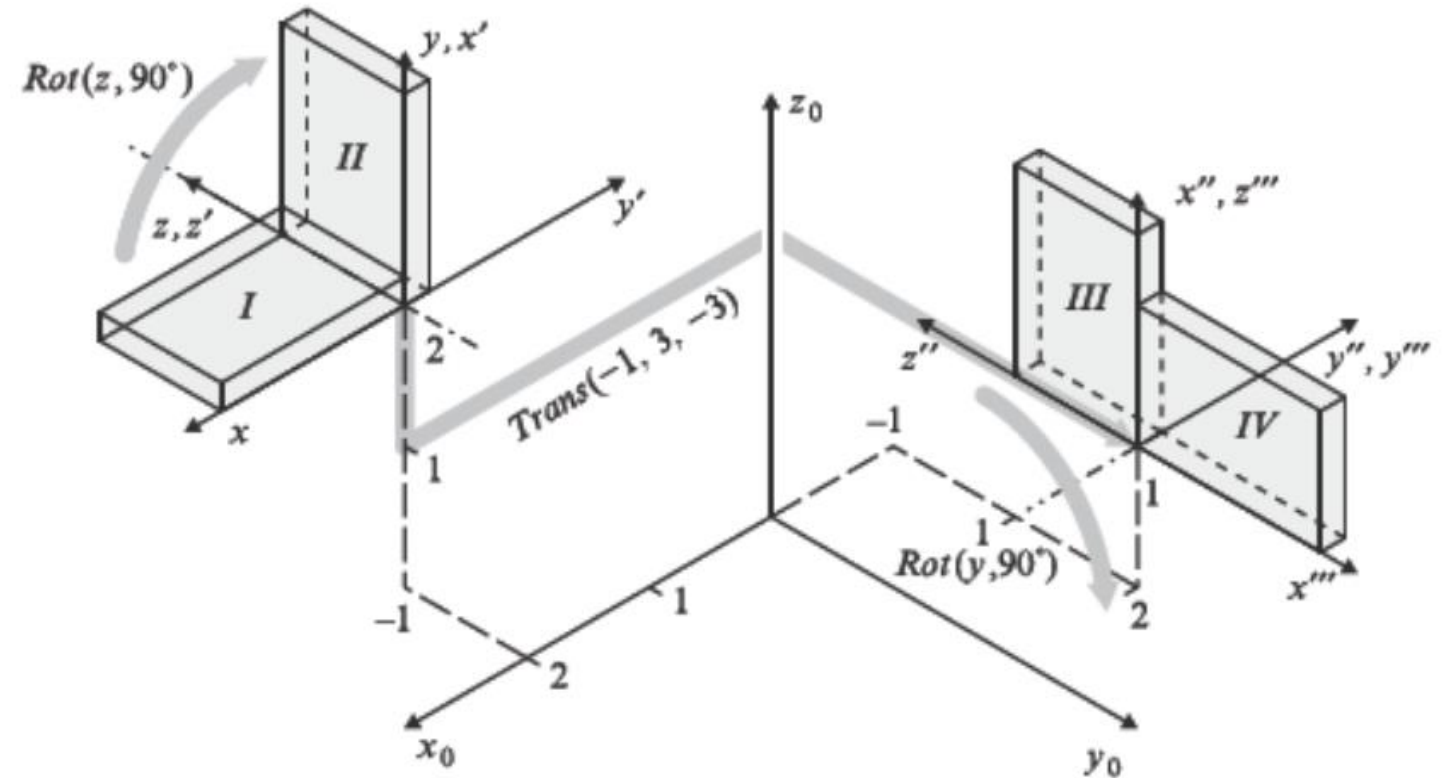


Let H introduced as the initial pose of the coordinate frame x, y, z

$$Rot(z, \gamma) = \begin{bmatrix} \cos \gamma & -\sin \gamma & 0 & 0 \\ \sin \gamma & \cos \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$Rot(z, 90) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



The displacement resulted in a new pose of the object and new frame x', y', z' is introduced.

Now, we shall displace this new frame for -1 along x' axis, 3 units along y' axis and -3 along z' .

$$\mathbf{H}_2 = \mathbf{H}_1 \cdot \text{Trans}(-1, 3, -3).$$

After translation a new pose of the object is obtained together with the new frame x', y' and z' . This frame will be finally rotated for 90° around y'' axis in the positive direction.

$$\mathbf{H}_3 = \mathbf{H}_2 \cdot \text{Rot}(y'', 90^\circ).$$

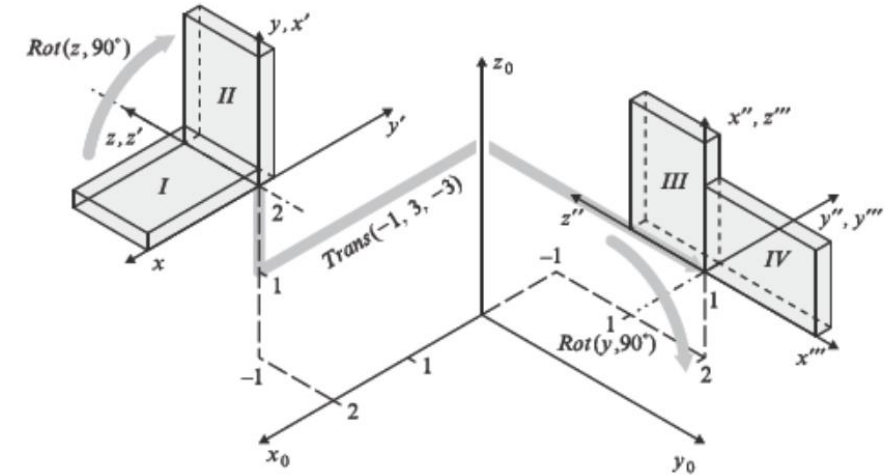
Or all together,

$$\mathbf{H}_3 = \mathbf{H} \cdot \text{Rot}(z, 90^\circ) \cdot \text{Trans}(-1, 3, -3) \cdot \text{Rot}(y'', 90^\circ) = \mathbf{H} \cdot \mathbf{D}.$$

$$\mathbf{D} = \text{Rot}(z, 90^\circ) \cdot \text{Trans}(-1, 3, -3) \cdot \text{Rot}(y'', 90^\circ)$$

$$= \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

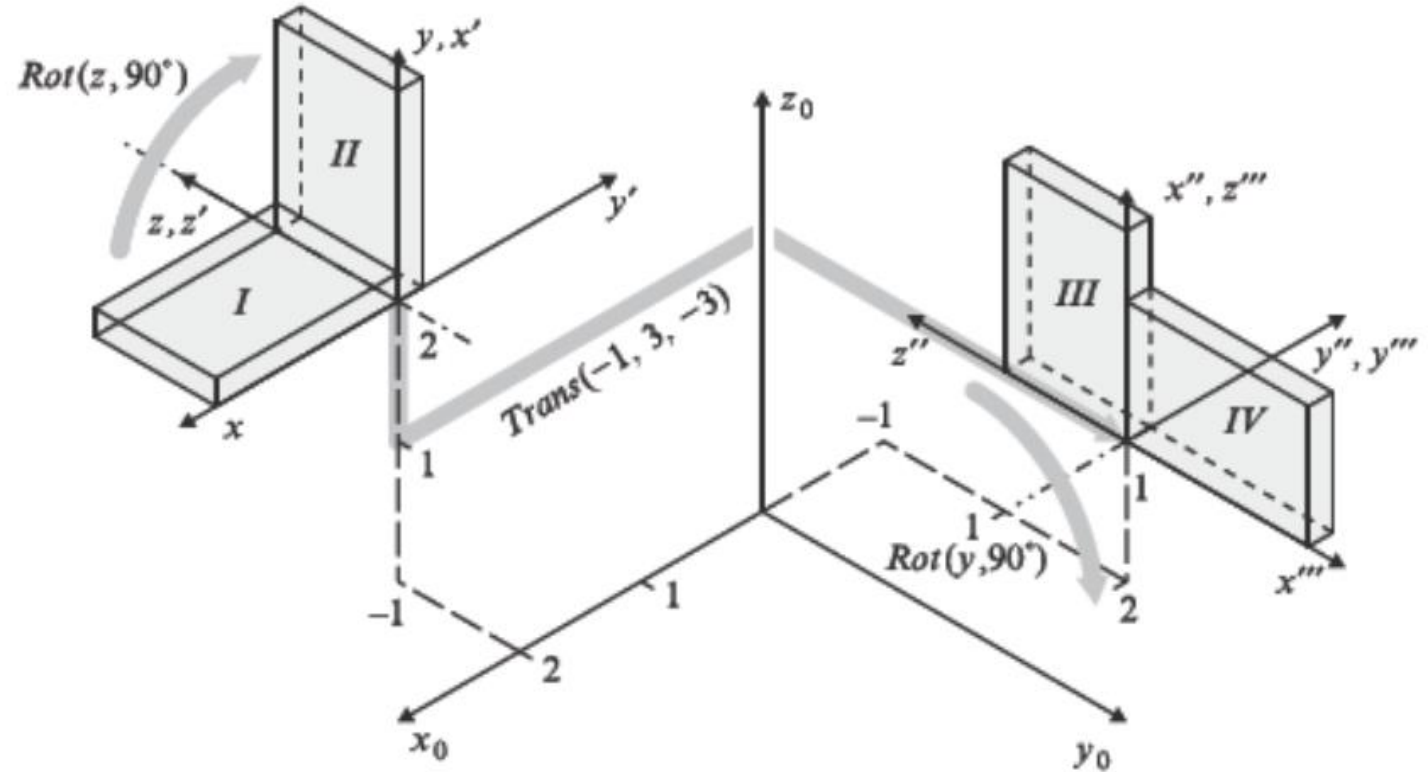
$$= \begin{bmatrix} 0 & -1 & 0 & -3 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$



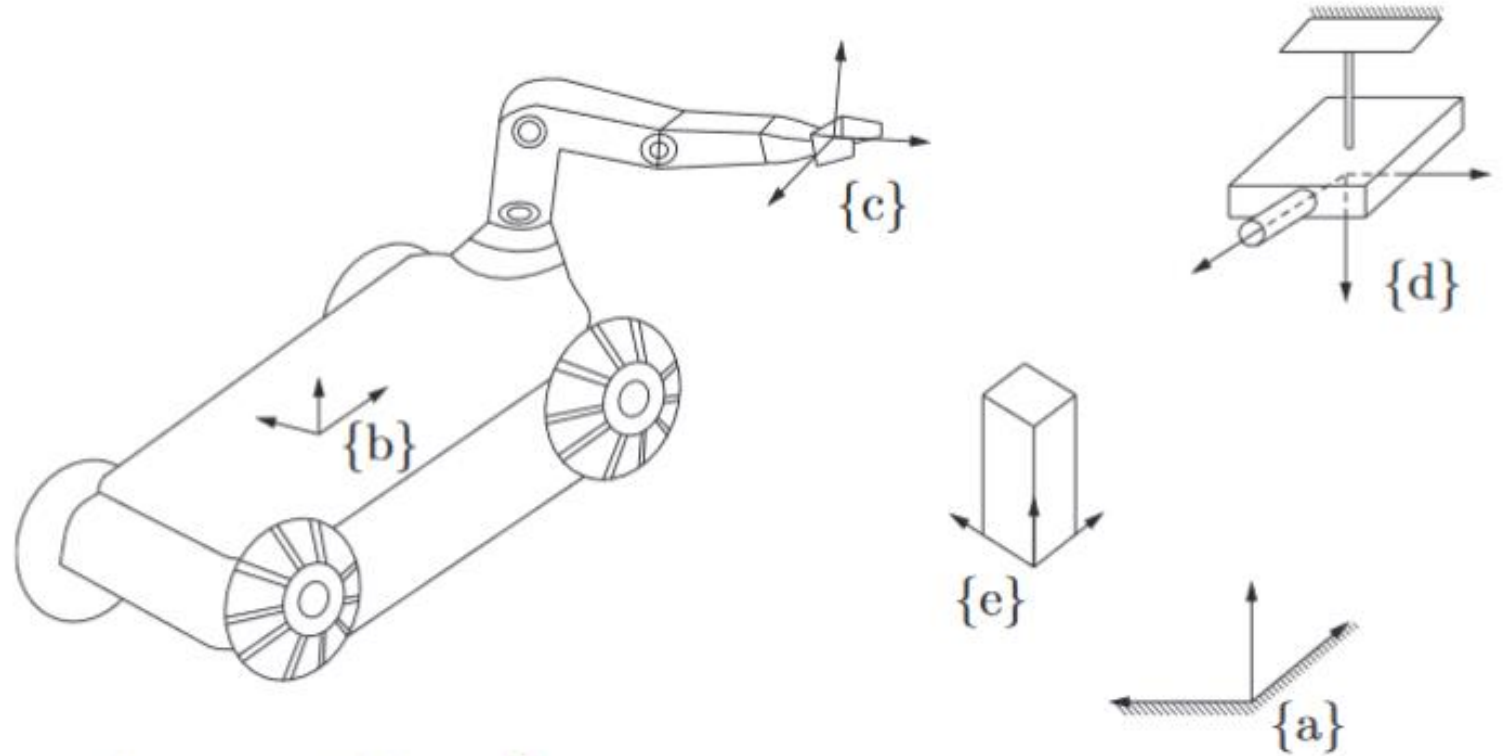
Example, cont.

$$\mathbf{H}_3 = \mathbf{H} \cdot \mathbf{D} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & -3 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{matrix} & x''' & y''' & z''' \\ \begin{bmatrix} 0 & -1 & 0 & -1 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} & \begin{matrix} x_0 \\ y_0 \\ z_0 \end{matrix} \end{matrix}$$



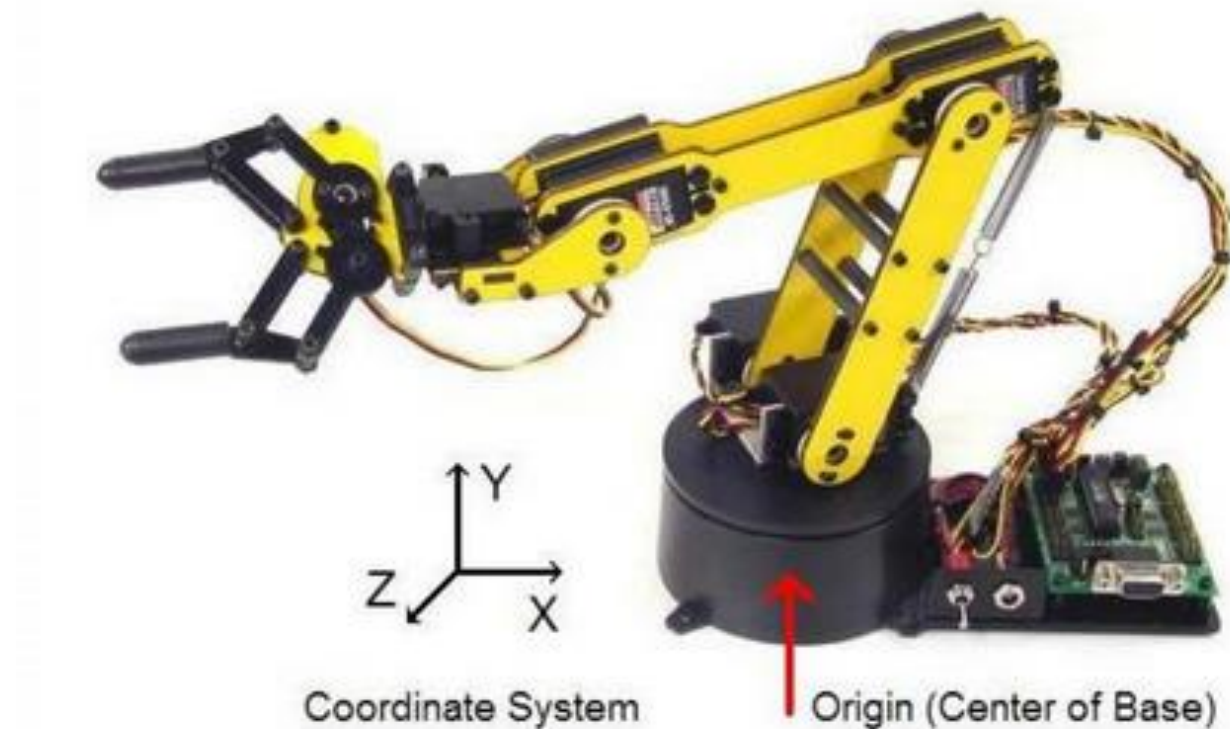
Example



$$T_{ce} = \begin{bmatrix} 0 & 0 & 1 & -75 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 & -260/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2} & 0 & 130/\sqrt{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

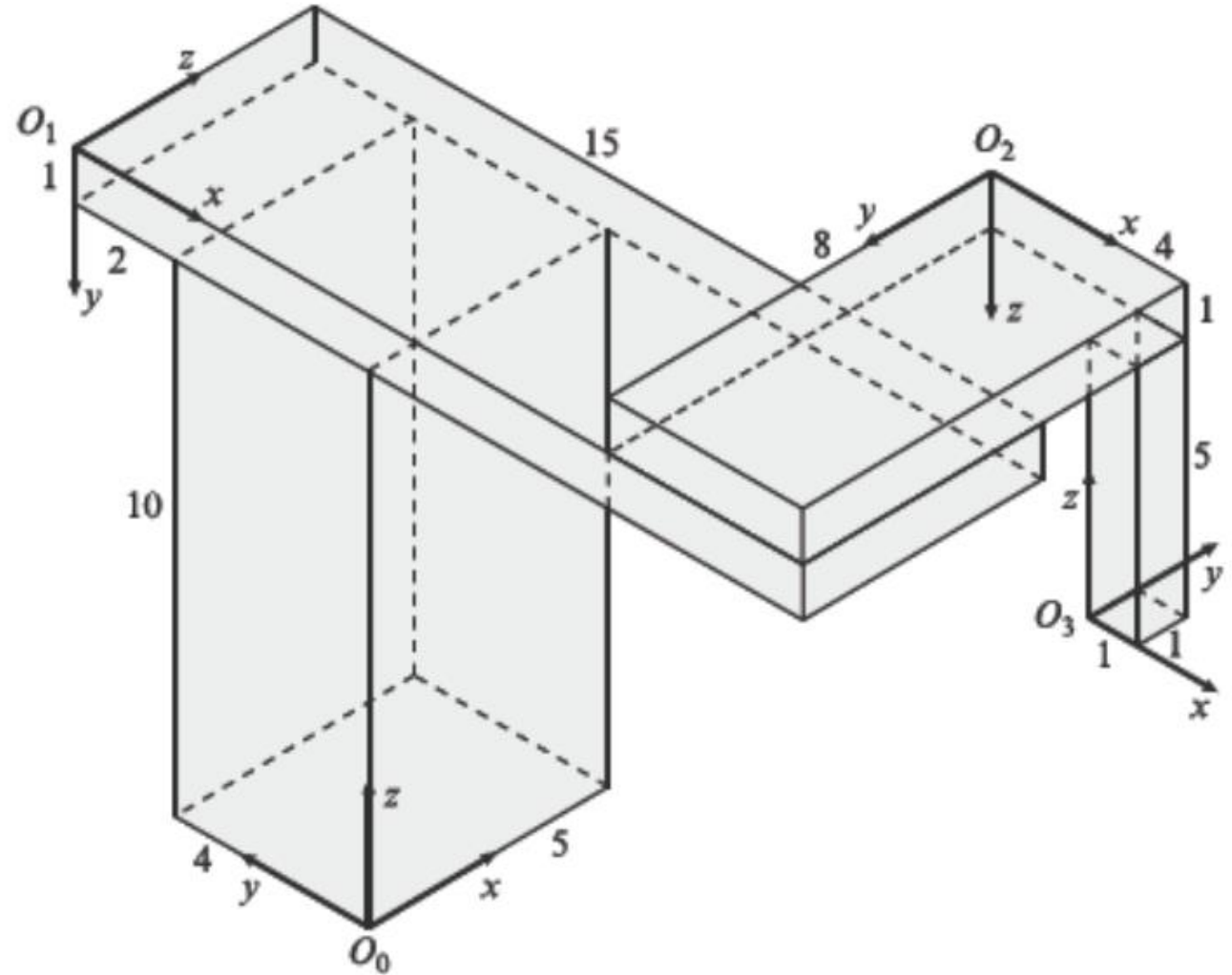
Geometrical Robot Model

A geometrical robot model is given by the description of the pose of the last segment of the robot (the end effector) expressed in reference (base) frame.



Example: Four block mechanical assembly

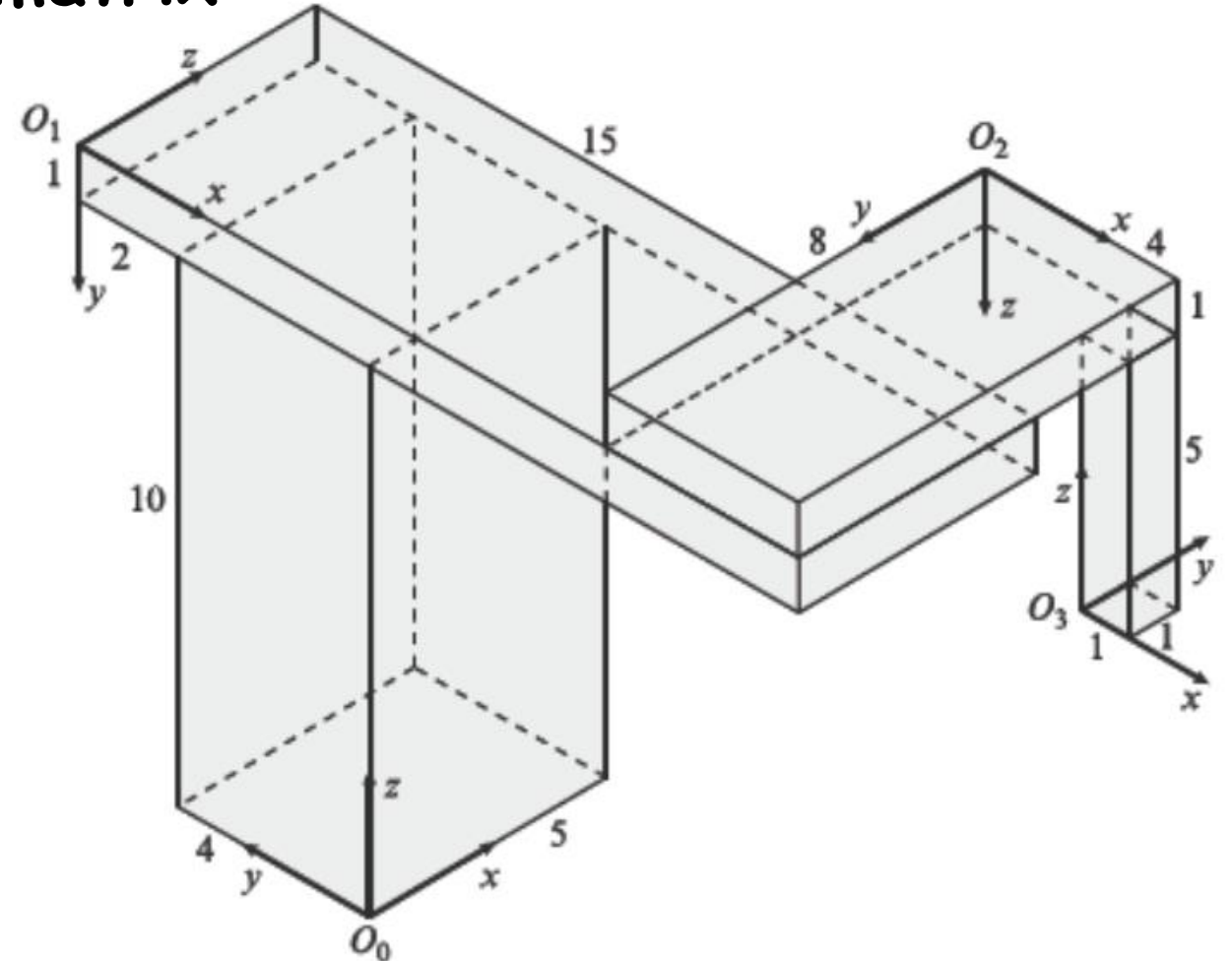
In this assembly, a frame is attached to each of the four blocks. The task is to calculate the pose of O_3 frame with respect to the reference frame O_0 .



The pose of the fourth block can be written with respect to the first one by the following matrix:

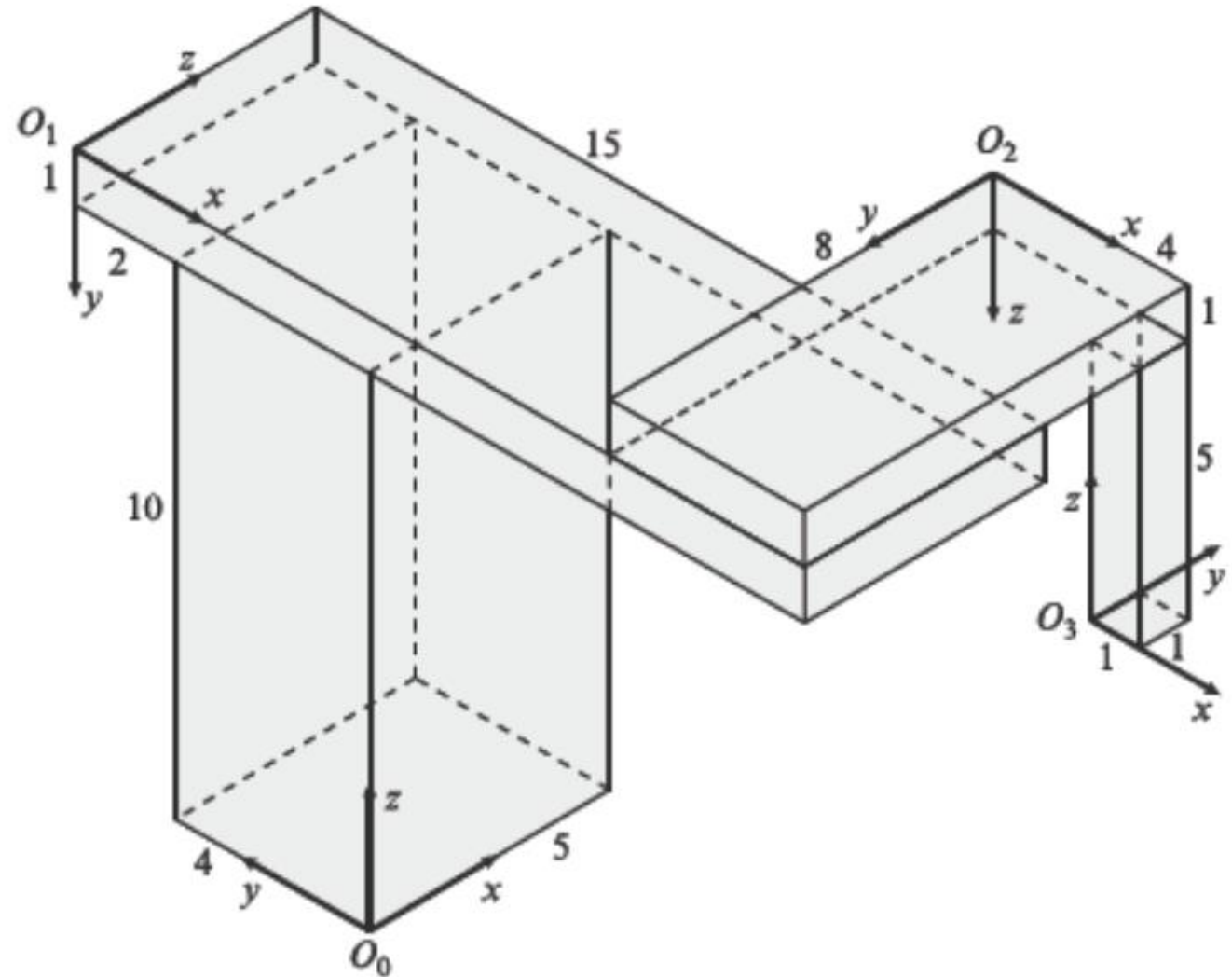
$${}^0\mathbf{H}_3 = {}^0\mathbf{H}_1 {}^1\mathbf{H}_2 {}^2\mathbf{H}_3.$$

$${}^0\mathbf{H}_1 = \left[\begin{array}{ccc|c} & \overbrace{\begin{matrix} x & y & z \end{matrix}}^{O_1} & & \\ \hline 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 6 \\ 0 & -1 & 0 & 11 \\ 0 & 0 & 0 & 1 \end{array} \right] \left. \begin{matrix} x \\ y \\ z \end{matrix} \right\} O_0.$$



$${}^1\mathbf{H}_2 = \begin{bmatrix} 1 & 0 & 0 & 11 \\ 0 & 0 & 1 & -1 \\ 0 & -1 & 0 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

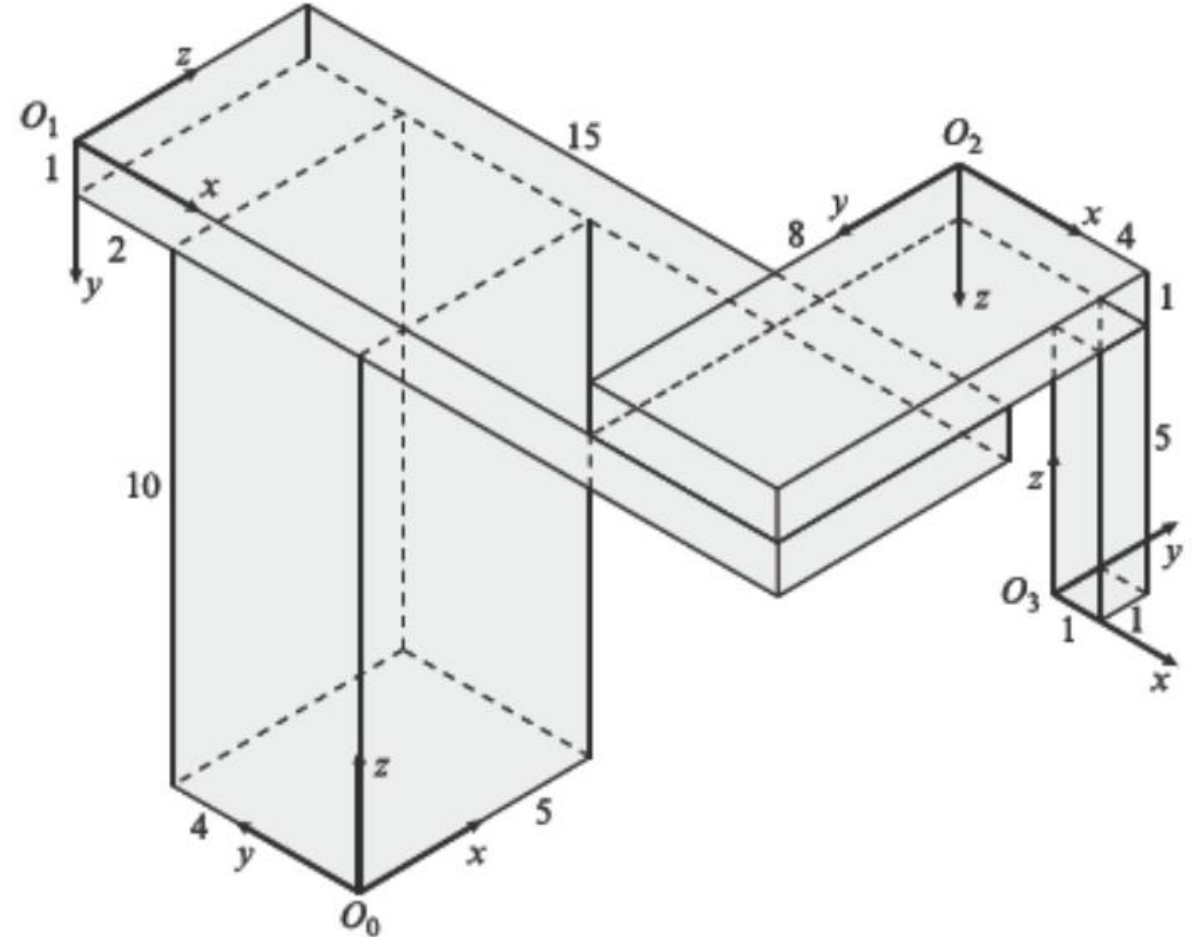
$${}^2\mathbf{H}_3 = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



The position and orientation of the fourth block with respect to the first one is given by:

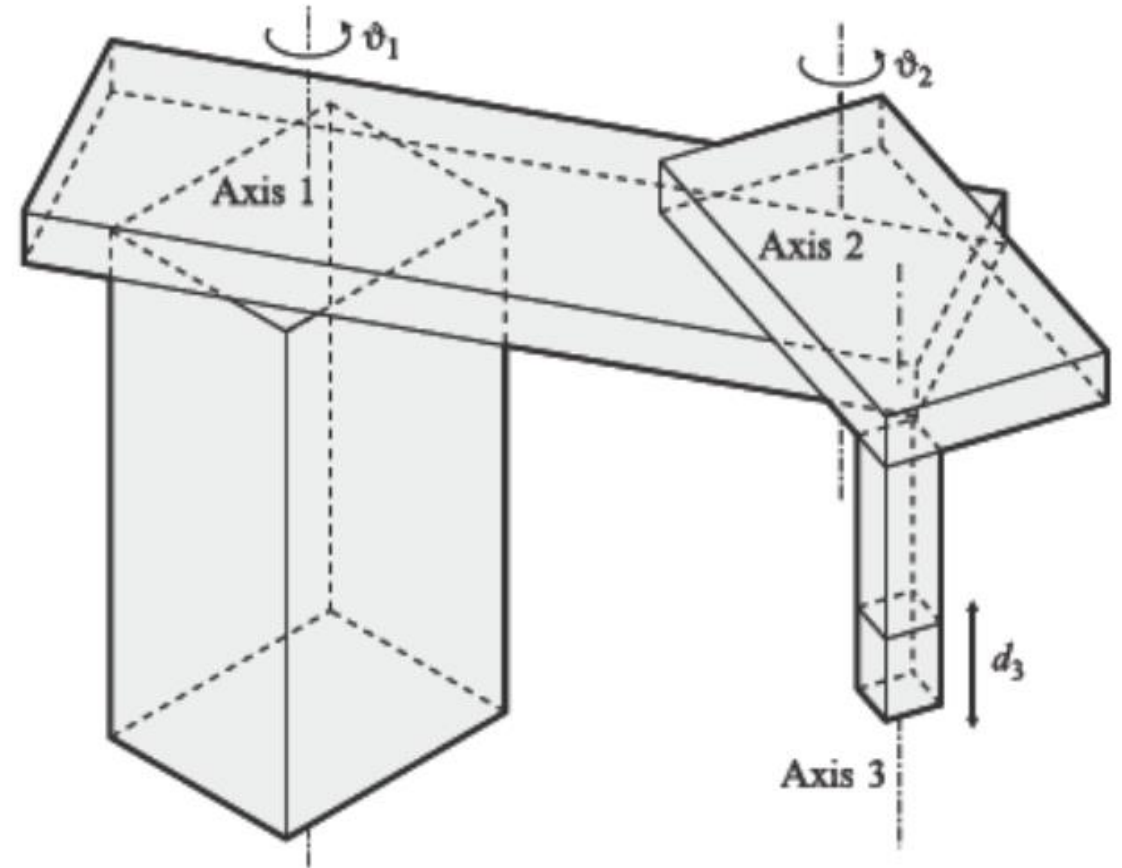
$${}^0\mathbf{H}_3 = \begin{bmatrix} 0 & 1 & 0 & 7 \\ -1 & 0 & 0 & -8 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The fourth column of the matrix represents the position of the origin of the O_3 frame with respect to the reference frame O_0 .



Remark:

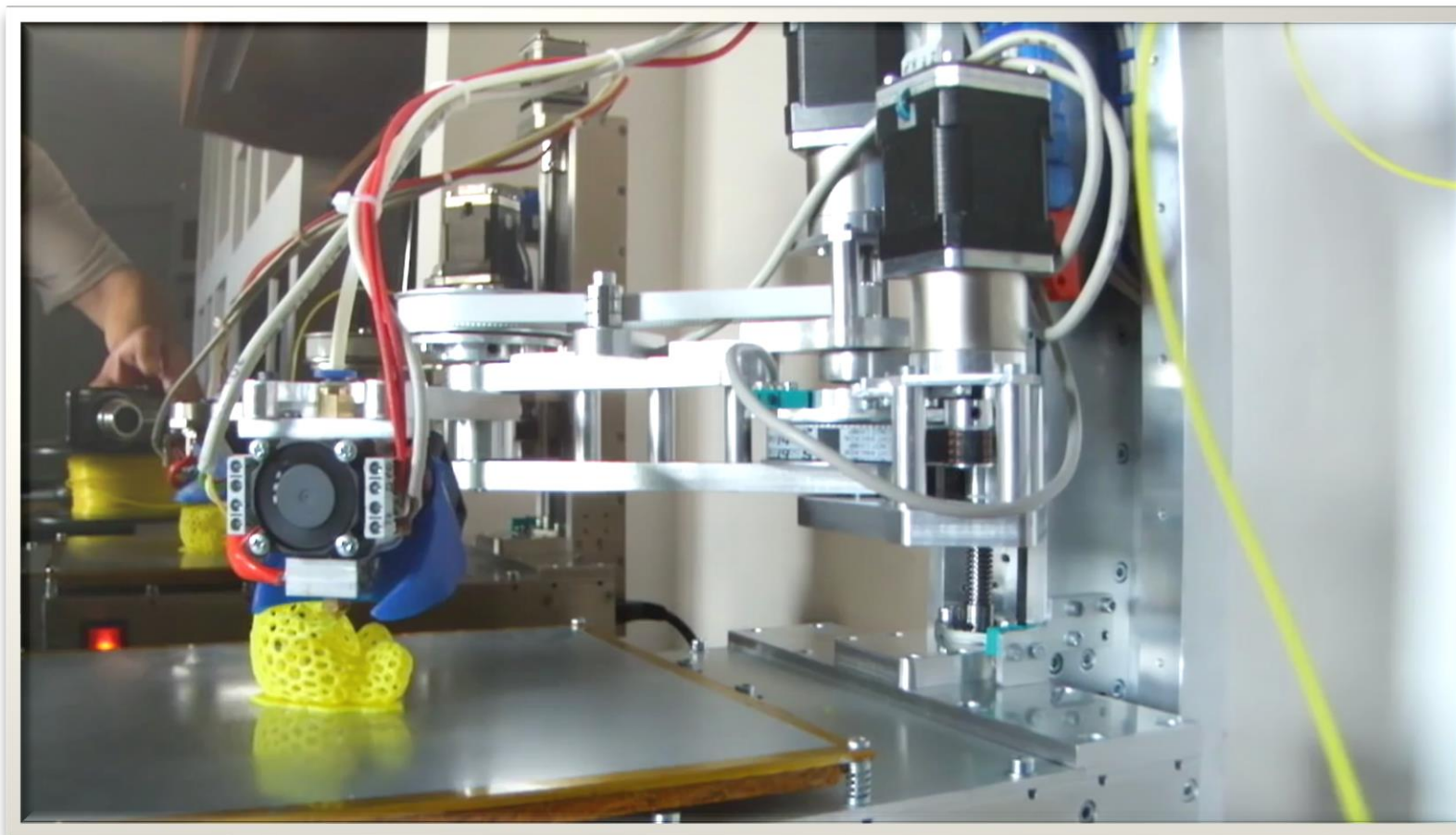
Consider that the first horizontal Plate rotates with respect to the first vertical block around axis 1 for angle θ_1 . The second plate also rotates around the vertical axis 2 for angle θ_2 . The last block is elongated for distance d_3 along the third axis. This is what is called SCARA robot.



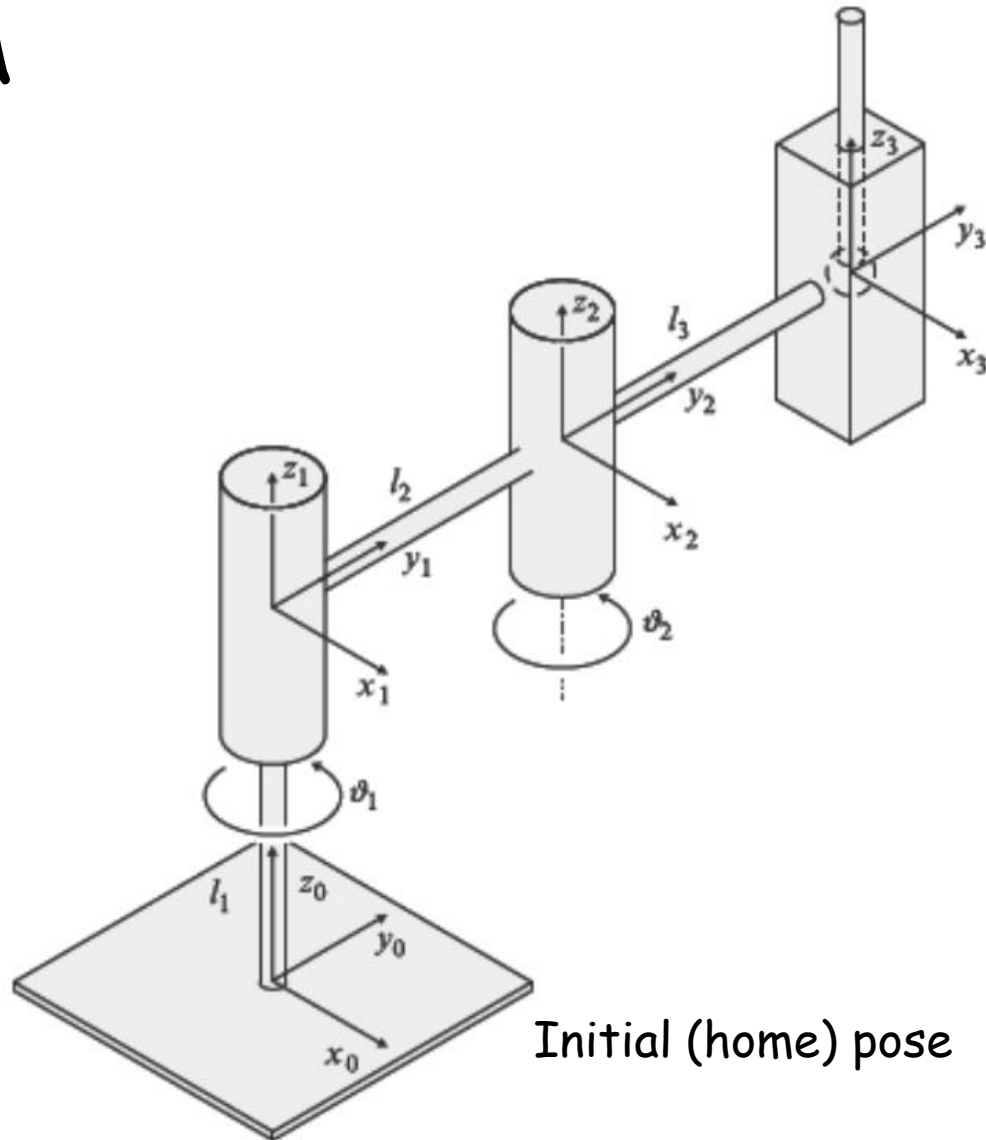
SCARA (read <https://en.wikipedia.org/wiki/SCARA>)



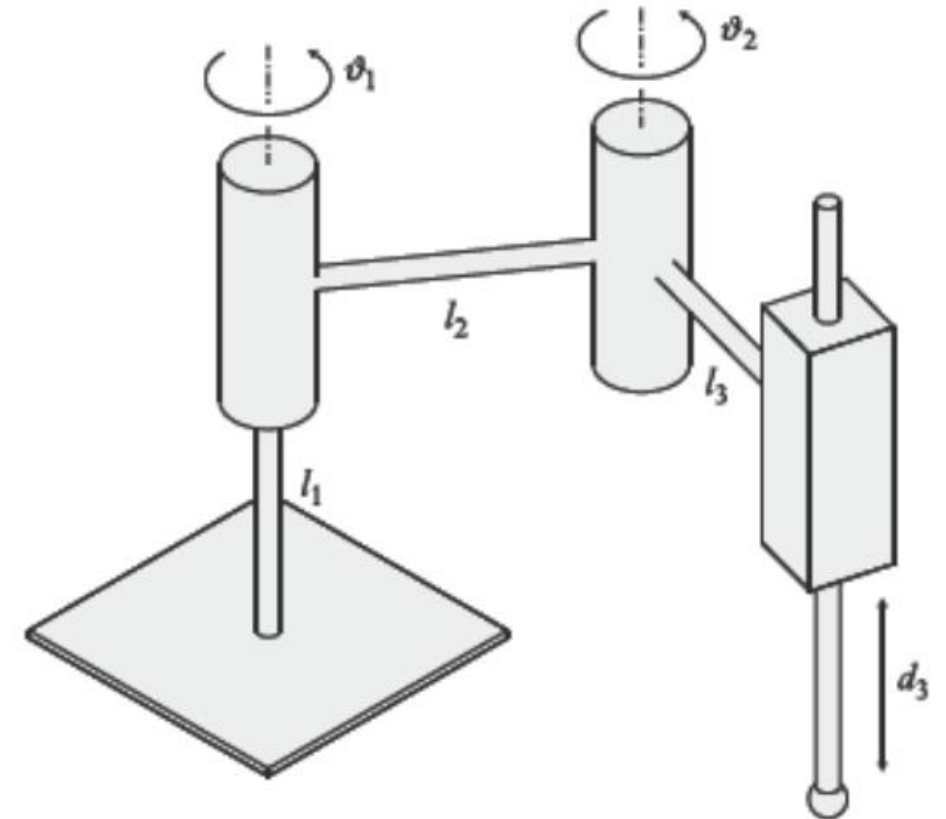
SCARA(application example)



SCARA



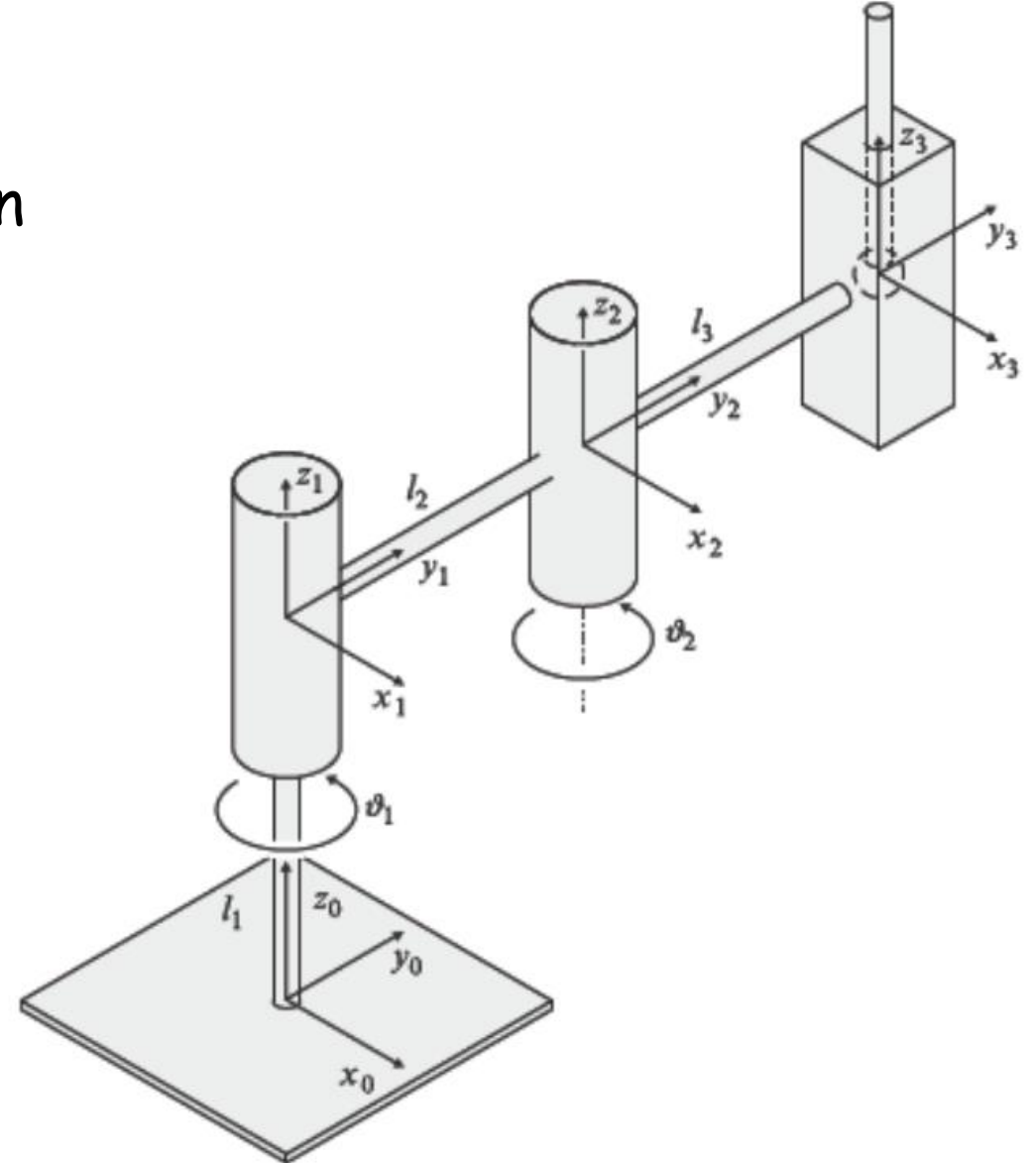
Initial (home) pose



Arbitrary pose

The pose of the end frame x_3, y_3, z_3 with respect to the reference frame x_0, y_0, z_0 is expressed by the following post-multiplication of three pairs of homogenous transformation matrices:

$${}^0\mathbf{H}_3 = ({}^0\mathbf{H}_1\mathbf{D}_1) \cdot ({}^1\mathbf{H}_2\mathbf{D}_2) \cdot ({}^2\mathbf{H}_3\mathbf{D}_3).$$



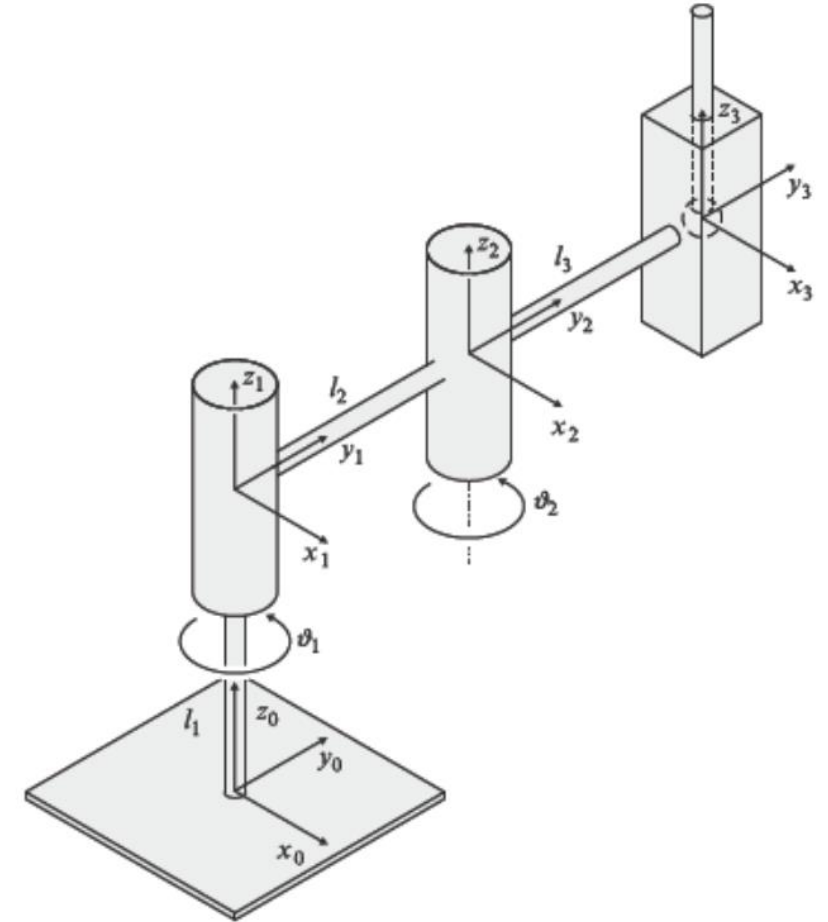
$${}^0\mathbf{H}_1\mathbf{D}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c1 & -s1 & 0 & 0 \\ s1 & c1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c1 & -s1 & 0 & 0 \\ s1 & c1 & 0 & 0 \\ 0 & 0 & 1 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1\mathbf{H}_2\mathbf{D}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & l_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c2 & -s2 & 0 & 0 \\ s2 & c2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c2 & -s2 & 0 & 0 \\ s2 & c2 & 0 & l_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2\mathbf{H}_3\mathbf{D}_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & l_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & l_3 \\ 0 & 0 & 1 & -d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0\mathbf{H}_3 = \begin{bmatrix} c12 & -s12 & 0 & -l_3s12 - l_2s1 \\ s12 & c12 & 0 & l_3c12 + l_2c1 \\ 0 & 0 & 1 & l_1 - d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$c12 = \cos(\vartheta_1 + \vartheta_2) = c1c2 - s1s2 \text{ and } s12 = \sin(\vartheta_1 + \vartheta_2) = s1c2 + c1s2.$$



Reading list

Kevin M. Lynch and Frank C. Park, 2017, *Modern Robotics*, 1st Edition, Cambridge University Press, chapter 3.

Questions?